“FIT & LIFT HUB”

A Project Report
Submitted in partial fulfilment of the
Requirement for the award of the Degree of

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)
By

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4019208

Under the esteemed guidance of

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DEPARTMENT OF INFORMATION TECHNOLOGY

BUNTS SANGHA’S S.M. SHETTY COLLEGE OF SCIENCE, COMMERCE & MANAGEMENT STUDIES

***(Affiliated to University of Mumbai)* MUMBAI,400076**
MAHARASHTRA
2022-23

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(Affiliated to University of Mumbai)

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-Amar Panigrahi

DECLARATION

I here by declare that the project entitled, **"FIT & LIFT HUB"** done at **Mumbai**, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)** to be submitted as final semester project as part of our curriculum.

Name and Signature of the Student

Student Detail

Name: Amar Panigrahi

Roll No: 125

Class: T.Y.B.Sc. I.T.

Project Detail

Project Topic: FIT & LIFT HUB

Frontend: HTML/CSS/JAVASCRIPT/PHP
Backend:

DBMS: MySQL
Project type: Website

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CHAPTER 1

INTRODUCTION

1.Introduction

- This assignment is designed to facilitate a gymming & fitness center to automatic it's operations of preserving facts and store them in shape of a large and person-pleasant database further facilitating easy access to the personnel.
- The aim of the project is to elaboration an online website.
- This **is** a web-based system that is built using PHP, CSS,JavaScript and MySQL database.
- The system is an online web application which can access by proper login.

1.1 Background

- Fitness is very important in our day-to-day life .to be fit we have to do some exercise and follow some diet plan.
- Earlier the gym proprietor has to maintain the file in manually manner which may be very hard for the individual.
- In this task Gym control system, it's miles on line website. Now a day we stay in age of information communication and technology. Have been maximum the works relying on internet site. To Present the statistics to the people we will conscious the people with the aid of developing this challenge.
- Gym management system help to the gym owner manage all data is easily.
- The website has the information of all trainer and the record of per—work and post-work of the mumbers about their daily basis.
- This website can help you in many ways.

1.2 Objectives

- The main objective of this website is to make the owner of the gym the easier way to keep record about members.
- It reduces the weight of the staffs to shop the information in manual
- Report No paper work.
- If there is any change in the record, we can it in a system which is very difficult to change in record file.
- His internet site has a record of every patron approximately his per-work and publish-paintings.
- In this the admin can create the groups of the members.
- Admin can add the members in the groups, And it will be very easy for the user to check the reports of financial and the body progress of the every individual.
- The goal of this system is reducing the work load of the owner so that they don't have a load of the paper work.

1.3 Purpose, Scope and Applicability

1.3.1 Purpose

- The intension of this is to create an automated system.it reduce the complexity and it can help in speeding the task.
- In present they keep the record in written manner sometime which leads to be very confusing.
- If there is any difference in the records, we can change it in the system which is not so easy in manual record file.
- This website can help us in many ways. It supports in storing the file.
- The user can check their record by themselves.
- Admin can change the record of clients.
- The aim of this is to make a technology is very easy format.

1.3.2 Scope

- The scope for this website is the admin can add the members.
- Admin can create a group where he/she can update the diets plans.
- Admin can maintain the attendance in the system which is a simple way than recording in a records file.
- The user can check their details. It is really very difficult to keep the record manually.
- It can also help the customer to register their name by not going to the gym location.
- They can get information about their workout by this website.
- They can check their pre-workout as well as post-work records by their own.
- The scope is admin can notice the customers profile and data. The admin can only change the records in the data. Admin can share anything about the gym.

1.3.3 Applicability

- This website could be very applicable for the person. They can register their call in this website. They should join their name the usage of username and password. This can beneficial in numerous approaches. Admin can create a group wherein he/she will be able to replace the data as for the fitness center. It's far applicable to save some information approximately the health club. No want of manual access. No need of carbon prints. The economic document also can store within the gadget very thoroughly. Admin can update the records of the customer. Admin also can replace the diet regime for patron as opposed to instructing on a workout location.

1.4 Organization of Report

- Survey of Technologies: This will give a describe description of the technologies that are available for the project and the ones that are being used for the current project.
- Requirements and Analysis: In this, we'll be taking the requirements from the user and use it to make strategies to implement in the project.

CHAPTER 2

SURVEY OF TECHNOLOGY

2.1Existing System

- The gym is active manually. The present system is time-consuming and also it is high priced, because it involves a lot of paperwork.
- To manually handle the system was very effortful task. But now-a-days computerization made calm to work.
- To grow efficiency with decrease cost.
- To decrease the load of paper work.
- To save schedule management for recording details of each and every single member and employee.
- To bring on required reports easily.

2.2 Proposed System

- The proposed machine is distinctly secured, due to the fact for login the system it requires the username and password that's one-of-a-kind for each department consequently providing each branch a specific view of the client statistics.
- It presents extensive variety of sure criteria in every window the client is running for better and quicker solution.
- It continues record for all criteria and transaction

2.3 Hardware Requirements

- Operating System: - windows XP or higher version
- Processor: -Intel(R) Core.
- Installed Memory (RAM): - Minimum 2GB
- System Type: - 32-bit Operating System
- Disk space: - 80GB minimum.

2.4 Software Requirements

- Operating System: Windows XP, Windows 8, 10.
- Web Designing: Html, CSS,
- Technology: PHP
- Front End: Php
- Back End: My SQL

2.5 Front-end web development details

- HTML provides the primary shape of sites, that is stronger and corrected through other technologies like CSS and JavaScript.
- CSS is used to govern presentation, formatting, and format.
- JavaScript is used to govern the behavior of different factors.

HTML

HTML is at the middle of every internet web page, regardless the trouble of a domain or wide variety of technologies concerned. It's an essential skill for any net professional. It's the starting point for everybody gaining knowledge of the way to create content for the net. And, happily for us, it's fantastically clean to examine.

CSS

CSS stands for Cascading Style Sheets. This programming language dictates how the HTML elements of a internet site must simply appear at the frontend of the page.

JavaScript

JavaScript is a more complex language than HTML or CSS, and it wasn't launched in beta shape till 1995. Nowadays, JavaScript is supported via all modern internet browsers and is used on almost every website at the web for extra powerful and complex capability.

3.2.2 Connectivity (front end and Back end):

PHP is an amazing and popular language!

It is strong sufficient to be on the middle of the massive running a blog device at the net (Word Press)! It is deep sufficient to run the most important social network (Facebook)! It is also clean enough to be a beginner's first servers, side language!

- PHP is an acronym for "PHP: Hypertext Pre-processor".
- PHP is an extensively-used, opens, source scripting language.
- PHP scripts are performed on the server.
- PHP is free to down load and use.
- PHP documents can contain textual content, HTML, CSS, JavaScript, and PHP code.
- PHP code are finished on the server, and the result is lower back to the browser as undeniable HTML.
- With PHP you aren't restrained to output HTML. You can output pics, PDF files, or even flash movies. You can also output any textual content, which includes XHTML and XML.

CHAPTER 3

REQUIREMENT AND ANALYSIS

3.1 Problem Definition

● Defining a problem is one of the critical sports of the assignment. The objective is to explain due the commercial enterprise problem to be solved & thereby determined the volume of the brand-new structures.

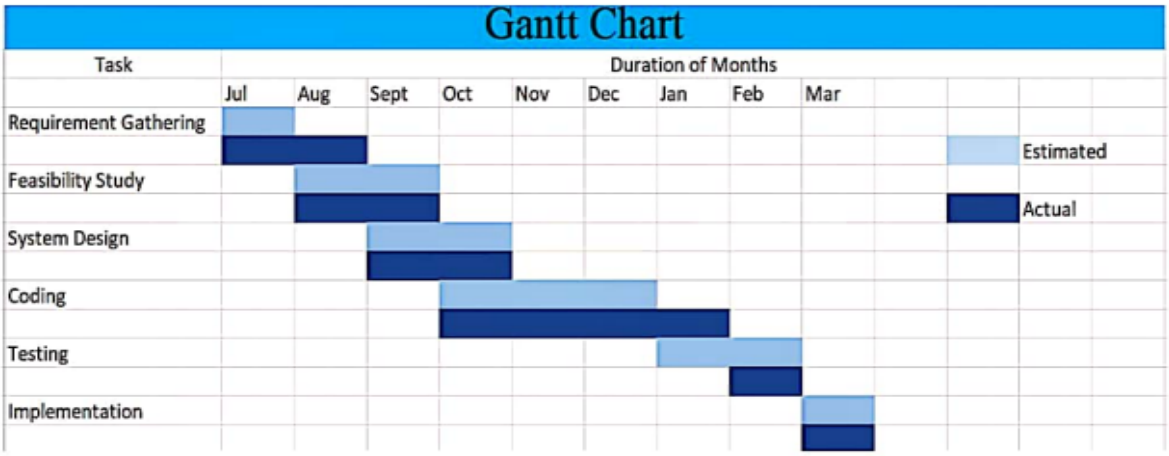
3.2 Requirement Specification

- Identifying the requirement necessarily involves specifying what the people who are going to use the system have in mind.
- The output of this stage is a set of accurate requirements, which are absolute and compatible.

3.3 Planning and Scheduling

3.3.1 Gantt Chart

It is a bar chart that represents a venture time table. It's, seem the start time and give up time of the elements of the assignment. It is used to display the present-day making plans popularity. Gantt charts incorporate helpful equipment for evaluating and planning of the complex projects They assist to plot out obligations that want to be finished. Help you discover a vital direction for a undertaking in which you must complete it on a selected date.



1.As seen in the Gantt chart above, the planning of the project started in the month of July. The expected time was a month for the arrival of the Requirement Specification stage but it took 2 months instead.

2. Feasibility study started in the month of august while the first stage was going on.

3. A rough device design become made within the subsequent stage, in which the diagrams are used for further growing the modules and placing the go with the flow of facts.

4. After the completion of previous stage, the coding stage was started where the design created in the previous step is converted into actual program.

5. The modules are tested individually and then integrated.

6. After integrating all the modules, the program will be tested as whole.

3.4 Hardware

1. Hardware Requirements: -

- Operating System: - windows XP or higher version
- Processor: -Intel(R) Core.
- Installed Memory (RAM): - Minimum 2GB
- System Type: - 32-bit Operating System
  Disk space: - 80GB minimum.

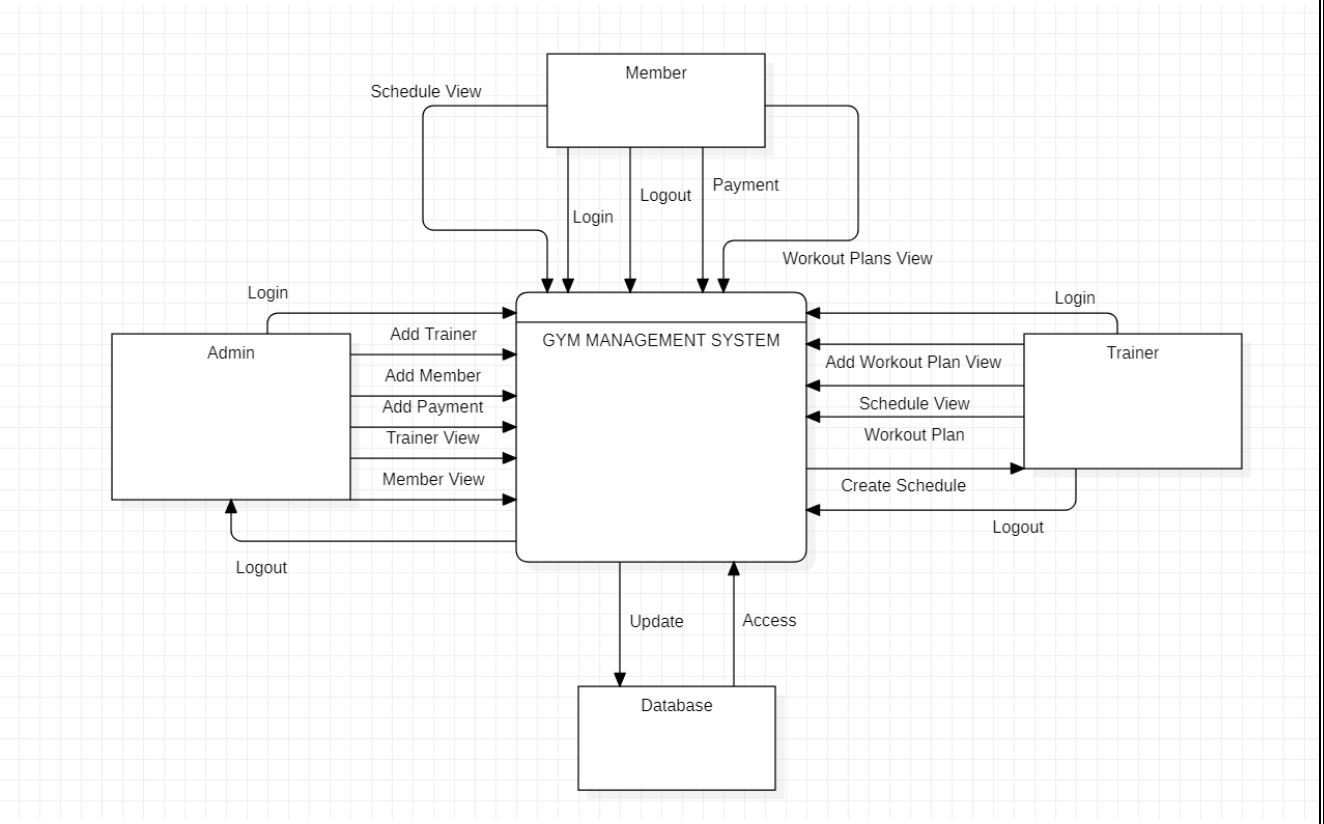
3.5 Software Requirements

- Operating System: Windows XP, Windows 8, 10.
- Web Designing: Html, CSS
- Technology: PHP
- Front End: php
- Back End: My SQL

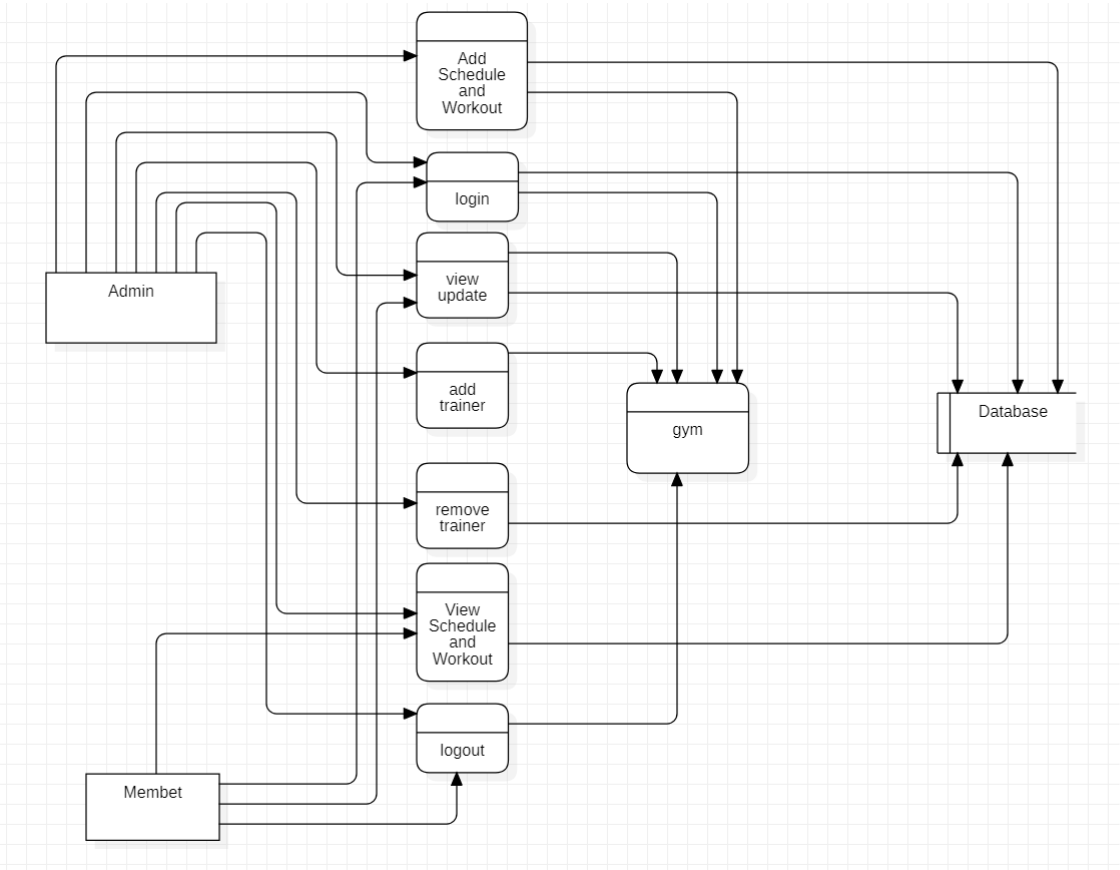
3.6 Conceptual Models

1. Data Flow Diagram Data-go with the flow-diagram (DFD) is used to sincerely illustrate the glide of facts in a business records system. DFD indicates the methods that are covered in a system to handover records from the input to the file garage and reviews technology

ZERO LEVEL

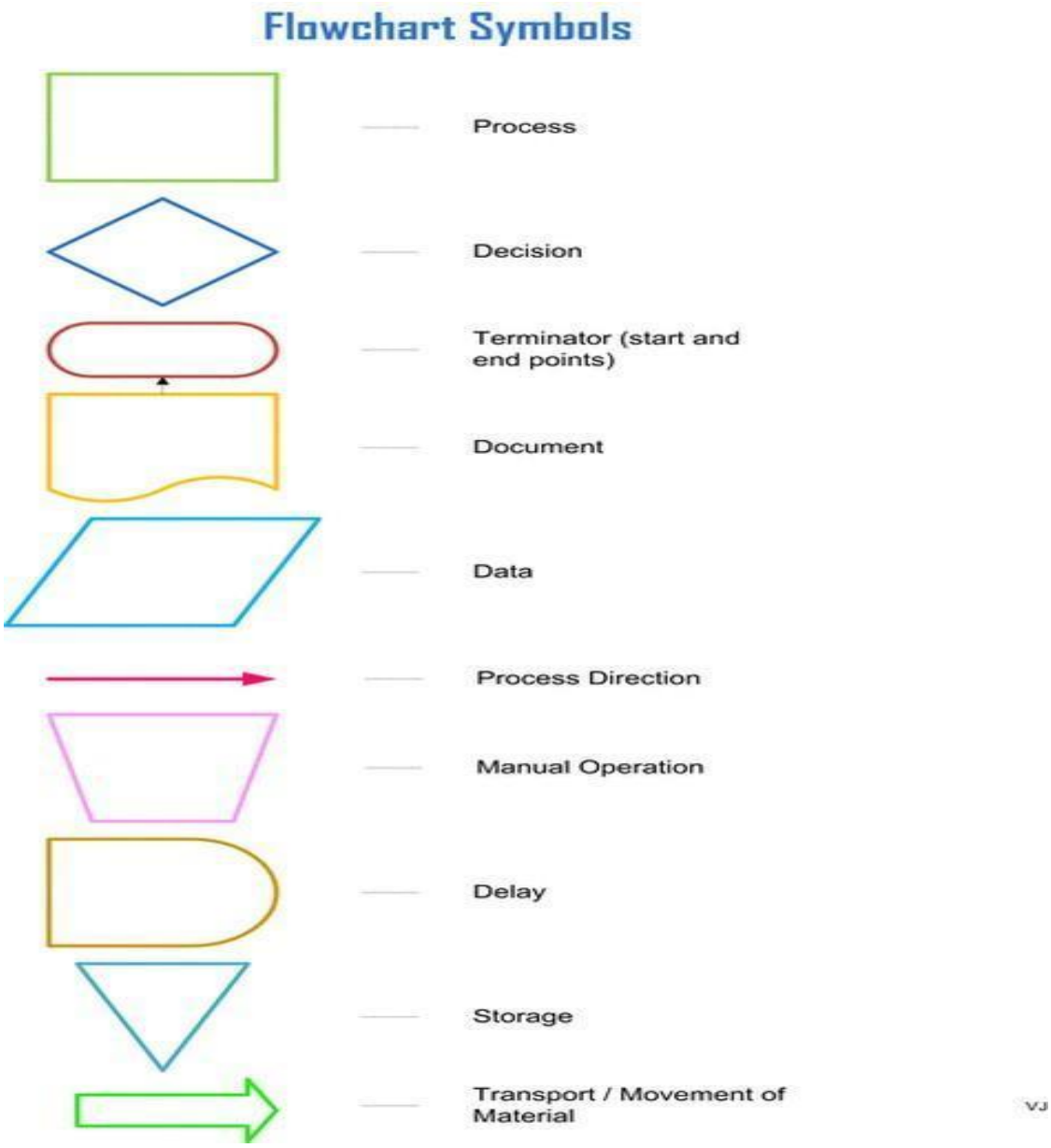


LEVEL 1



System Flowchart

A flowchart displays the depiction of steps. It indicates steps in modern order and is widely utilized in offering the waft of algorithms, workflow or tactics. Generally, a flowchart shows the stairs as bins of various sorts, and their order with the aid of linking them with arrows.



CHAPTER 4

SYSTEM DESIGN

4.1:-Modules:

Admin: Admin will manage the website like their security, privacy, database, update etc.

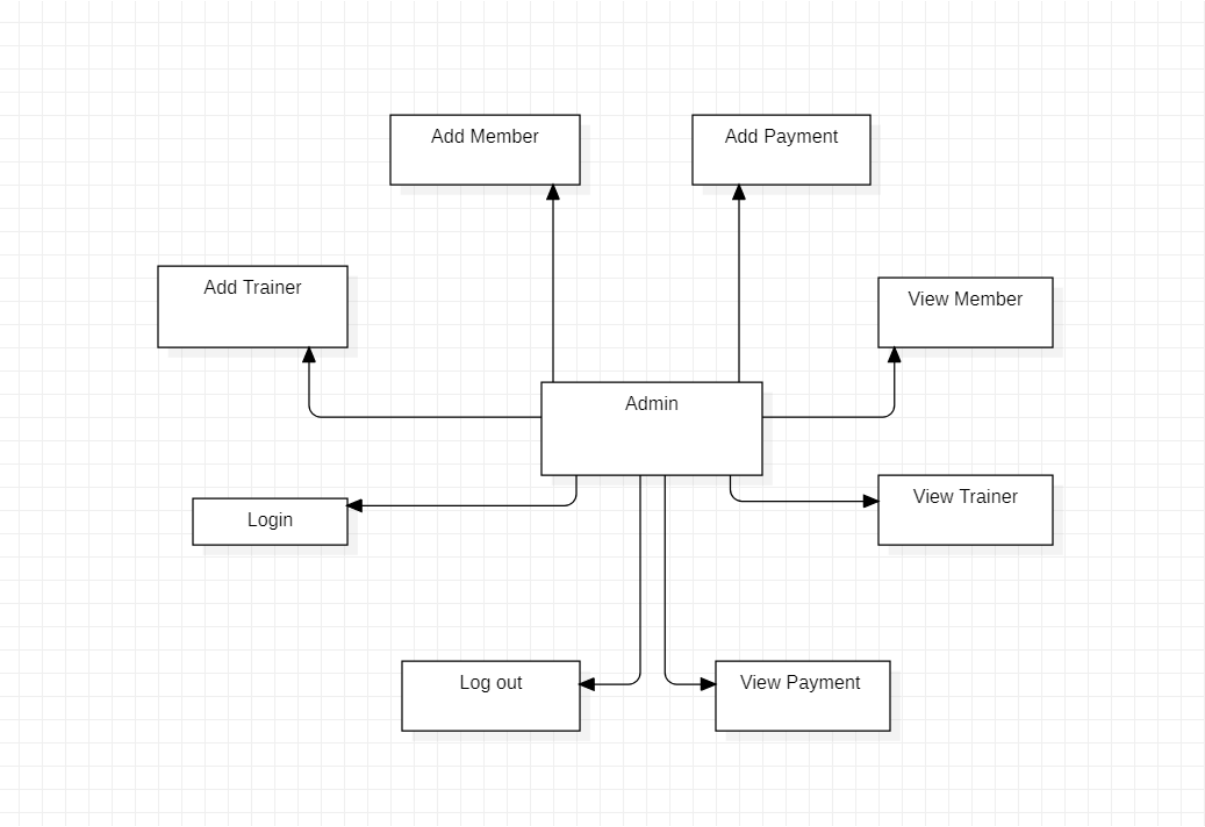
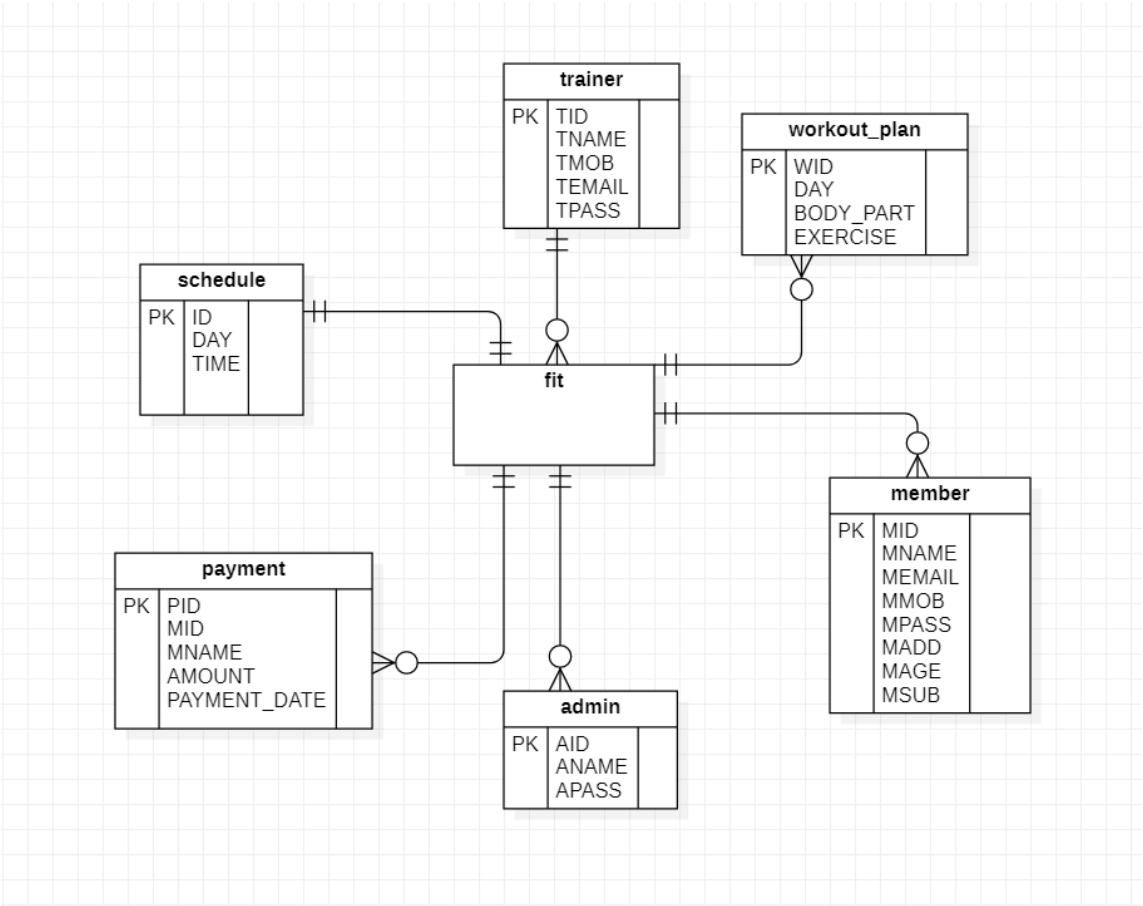


Diagram of Admin Module

- **Login:** Admin need to login by providing the login capability to access the below given admin module
- **Add Member:** Admin can add member.
- **View or Edit payment:** Admin Can view description of any payment and make changes into that.
- **View member:** Admin can see the member details.
- **Add Package:** Admin can add new package.
- **Logout:** Admin can logout from his account.



ER -DIAGRAM

ENTITY RELATIONSHIP DIAGRAM

- Entity–relationship model (ER model) in software engineering is a way to describe a database.
- The database is beginning with the relation database that is used to store the distinct statistics of training it h table layout.
- An ERD is a logical representation of an organisation's records, and consists of three primary additives:

Entities

- Entity is a real-world object.
- Each entity has its own properties or attributes.
- Entities are represented by rectangle.

Attributes

Attributes also are referred to as homes of entities.

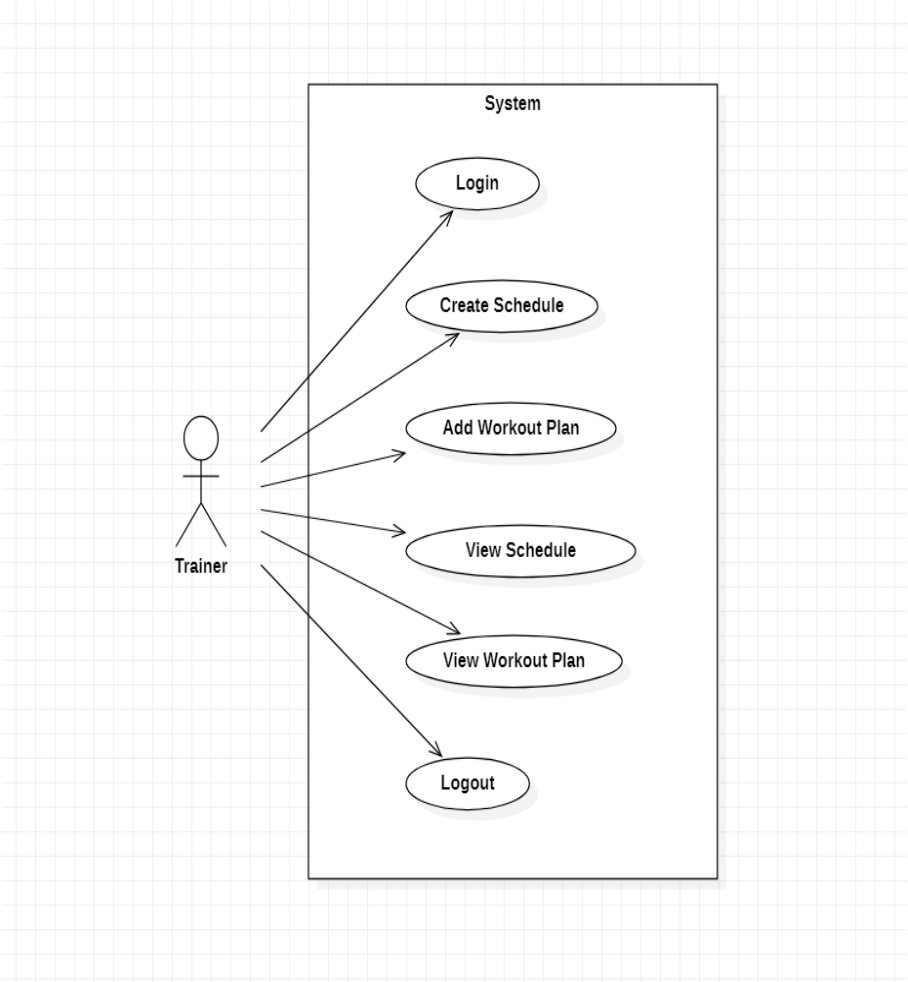
- When defining an attribute, an analyst should country why the attribute is critical.
- A sound know-how of an organization's business should help the analyst in compiling applicable attributes.

Relationships

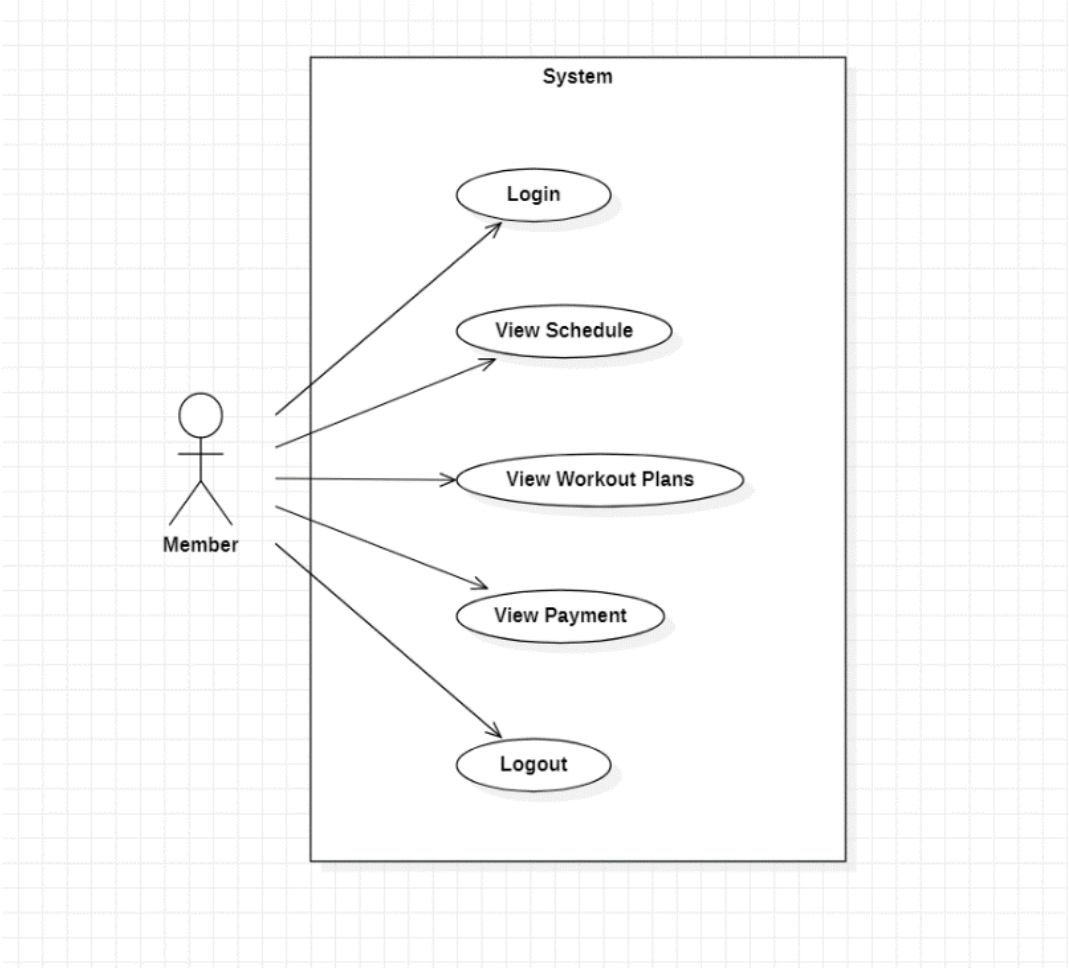
- Relationships between entities are represented by lines. Relationships link between two or more entities.
- There are three types of relationship degrees such as
    - Unary (between instances of one entity type)
    - Binary (between instances of two entity types)
    - Ternary (between three entity types).

Use Case Diagram

- **Actor: -** An actor portrays any entity that perform positive roles in systems. An actor in use case diagram interacts with the use case.
- **Use Case:** A use case is a visual illustration of enterprise functionality in a system each use case the collection of transaction completed by using the machine.
- **System boundary: -** A system boundary defines the scope of the what as system will be. A system cannot have infinite functionality. A system boundary defines the limits of system
- **Association: -**This is used to show the participation of actor in use case.



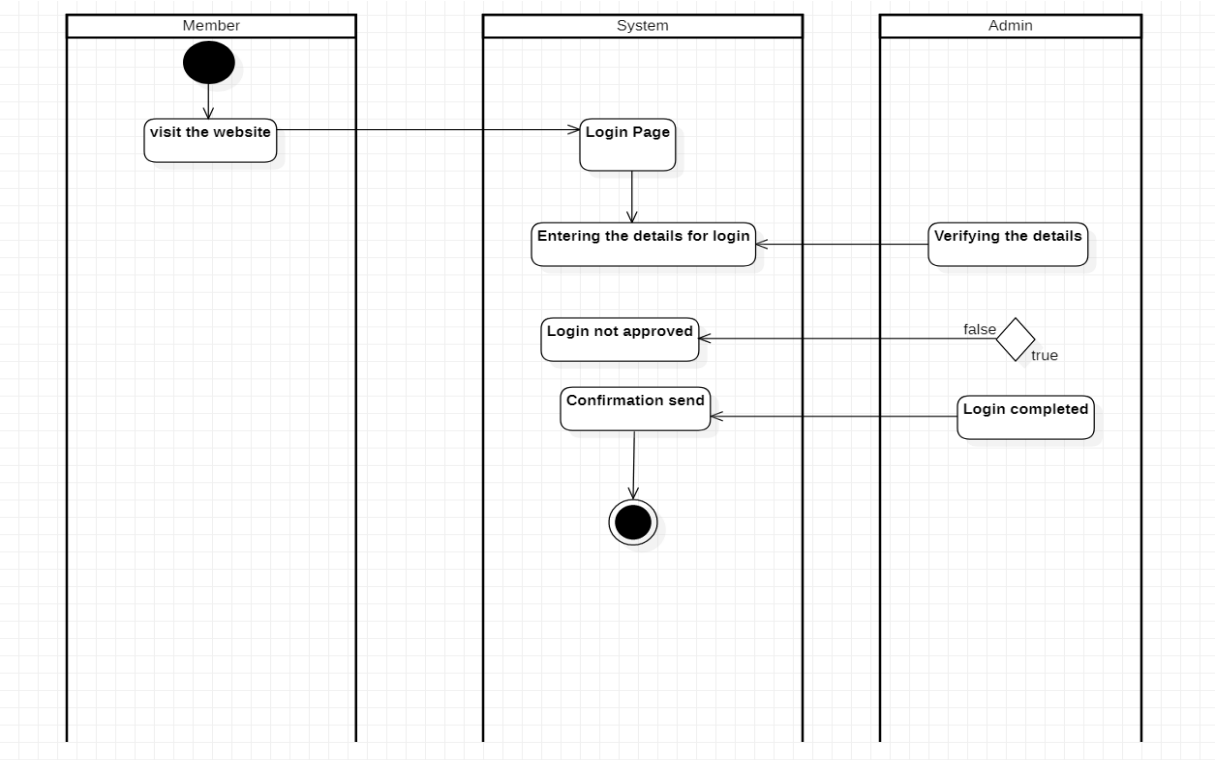
Trainer Usecase Diagram



Member Use case Diagram

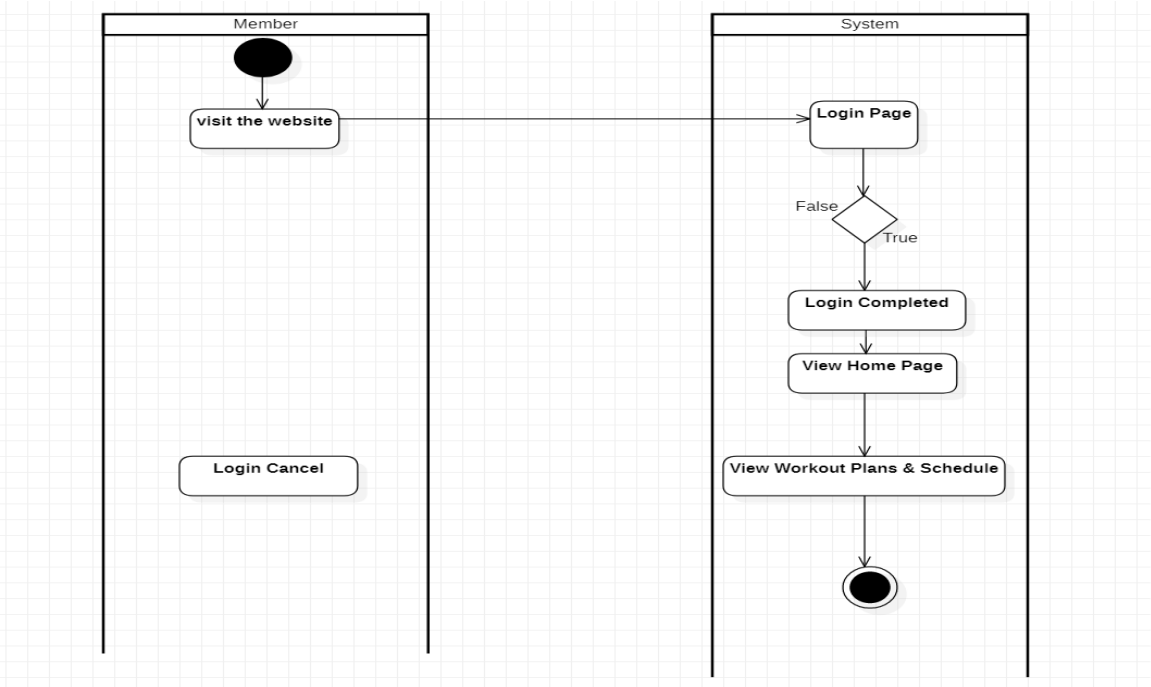
ACTIVITY DIAGRAM

Trainer Registration: -the activity is about the trainer registration. The trainer can register and can create a username and password. The process is the trainer has to view a website and fill all the details in the registration page with full name and address. If the details of the trainer are true then it will register his/ her name in the database



Member Diagram

The member can firstly visit the website and then register their name. The registration includes full name, address, id, email, phone no. And then the member should create username and a customized password for the login page.



4. Data Integrity and constraints:

Admin login

Field	Data type	Description	Constrain
U Id	Int	Login id of admin	Primay key
Pass word	Var char	Login password of admin	Not Null

Trainer Login

Field	Data type	Description	Constrain
U Id	Int	Login id of admin	Primar y key
Passwor d	Var char	Login password of admin	Not Null

4.2 User-Interface-Design:

User interface design is design between the user and the application. User interface is the initial interaction with the user. User interface elements should have the good user experience. User interface is used for investigating and evaluating user requirements. The user interface-design will help the user to use this application very. The principles of user interface designs are listed below.

- It should be very straightforward and easy to convenient.
- It should be very softly working
- It should be functional.
- It should be convenient for the user.
- It should be error prevented

4.3 Security Issues

- This website will have a security for all the users.
- Every user has their own username and password.
- The website has an authentication the user can pass.
- It is very important for the website to have a Security authentication.

4.4 Limitations:

- Every user should have an internet connection.
- The user should know how to use the computer or the website.
- The user should have an email account.

4.5 Data Design

4.5.1 Admin Login:

Field Name	Data type	Size	Null
AID	Varchar	30	Not Null
ANAME	Varchar	30	Not Null
APASS	Varchar	30	Not Null

Table 1 : admin

Admin Login:

1) In the admin login there is the email and password are needed.
2) In this email and password is having data type is varchar.
3) In this data size 30 that mean it will take 30 variable character.
4) In this on null table is also present which means value of the email and password can't be null.

4.5.2 Member:

Field Name	Data type	Size	Null
MID	Int	30	Primary Key
MNAME	Varchar	30	Not Null
MEMAIL	Varchar	30	Not Null
MMOB	Varchar	30	Not Null
MPASS	Varchar	30	Not Null
MADD	Varchar	30	Not Null
MAGE	Int	30	Not Null
MSUB	Varchar	30	Not Null

Table 2 : member

4.5.3 Payment:

Field Name	Data type	Size	Null
PID	Int	30	Primary Key
MID	Varchar	30	Foreign Key
MNAME	Int	30	Not Null
AMOUNT	Int	30	Not Null
PAYMENT_DATE	Date	6	Not Null

Table 3 : payment

4.5.4 Schedule:

Field Name	Datatype	Size	Null
ID	Varchar	30	Not Null
DAY	Int	30	Not Null
TIME	Int	30	Not Null

Table 4 : schedule

4.5.5 Trainer:

Field Name	Datatype	Size	Null
TID	Int	30	Foreign key
TNAME	Varchar	30	Not Null
TMOB	Int	30	Not Null
TEMAIL	Int	30	Not Null
TPASS	Int	30	Foreign key

Table 5 : trainer

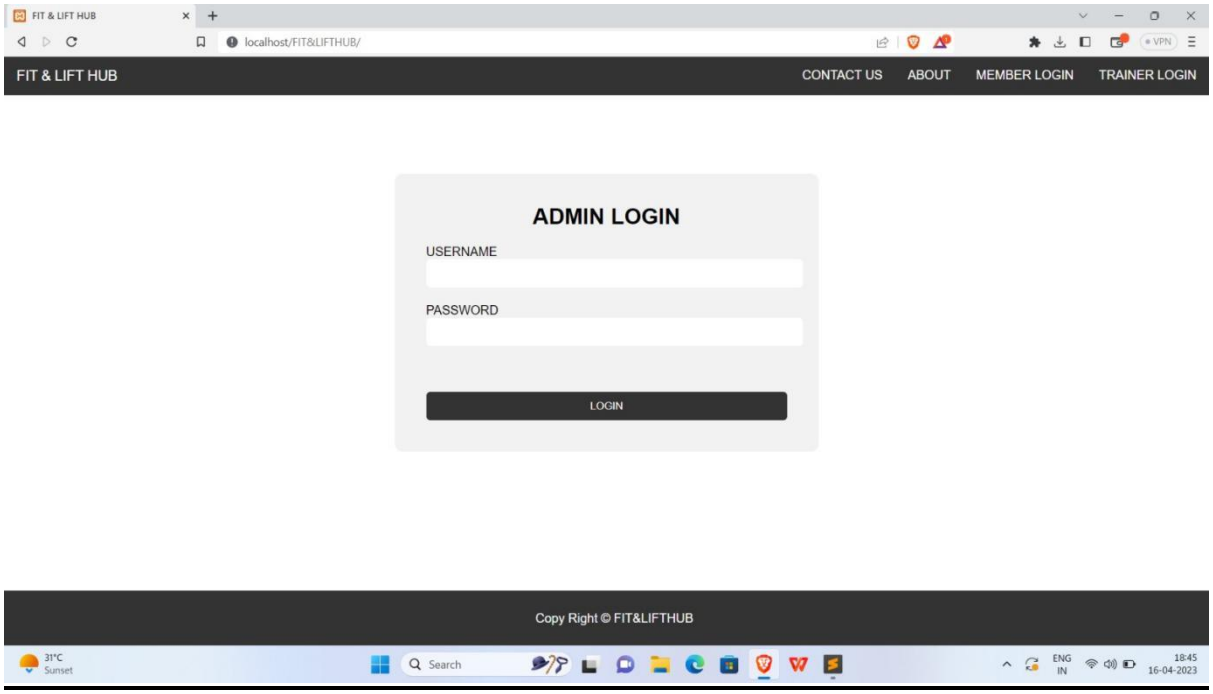
4.5.5 Workout_plan:

Field Name	Datatype	Size	Null
WID	Int	30	Foreign key
DAY	Varchar	30	Not Null
BODY_PART	Int	30	Not Null
EXERCISE	Int	30	Not Null

Table 6 : workout_plan

CHAPTER 5

RESULT & DISCUSSIONS



5.1 Admin login

index.php

```
index.php

<?php
include "database.php";
?>
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>FIT & LIFT HUB</title>
    <style>
body {
  font-family: Arial, sans-serif;
  margin: 0;
}
.back {
  background-image: url("images/background.jpg");
  background-repeat: no-repeat;
  background-size: cover;
}
nav {
  background-color: #333;
  overflow: hidden;
}
nav a.title {
  float: left;
  color: white;
  text-align: center;
  padding: 14px 16px;
  text-decoration: none;
  font-size: 18px;
}
nav a.title:hover {
  background-color: #ddd;
  color: black;
}
nav a {
  float: right;
  color: white;
  text-align: center;
  padding: 14px 16px;
  text-decoration: none;
  font-size: 16px;
}
nav a:hover {
  background-color: #ddd;
  color: black;
}
.login {
  margin-top: 100px;
  max-width: 500px;
  margin-left: auto;
  margin-right: auto;
  background-color: #f2f2f2;
  padding: 20px;
  border-radius: 10px;
```

index.php

```
56    }
57    .login .heading {
58      text-align: center;
59      font-size: 28px;
60      margin-bottom: 20px;
61    }
62    .login .log {
63      margin: 20px;
64    }
65    .login .input {
66      display: block;
67      width: 100%;
68      padding: 10px;
69      margin-bottom: 20px;
70      border: none;
71      border-radius: 5px;
72    }
73    .login .btn {
74      display: block;
75      width: 100%;
76      padding: 10px;
77      margin-top: 20px;
78      background-color: #333;
79      color: white;
80      border: none;
81      border-radius: 5px;
82      cursor: pointer;
83    }
84    .login .btn:hover {
85      background-color: #ddd;
86      color: black;
87    }
88    .login .error {
89      color: red;
90      margin-bottom: 10px;
91    }
92    .footer {
93      background-color: #333;
94      color: white;
95      text-align: center;
96      padding: 10px;
97      position: fixed;
98      bottom: 0;
99      width: 100%;
100   }
101       </style>
102   </head>
103   <body class="back">
104       <nav>
105           <a class="title" href="index.php">FIT & LIFT HUB</a>
106           <a href="trainer_login.php">TRAINER LOGIN</a>
107           <a href="member_login.php">MEMBER LOGIN</a>
108           <a href="aboutus.php">ABOUT</a>
109           <a href="contactus.php">CONTACT US</a>
110       </nav>
```

Search

index.php

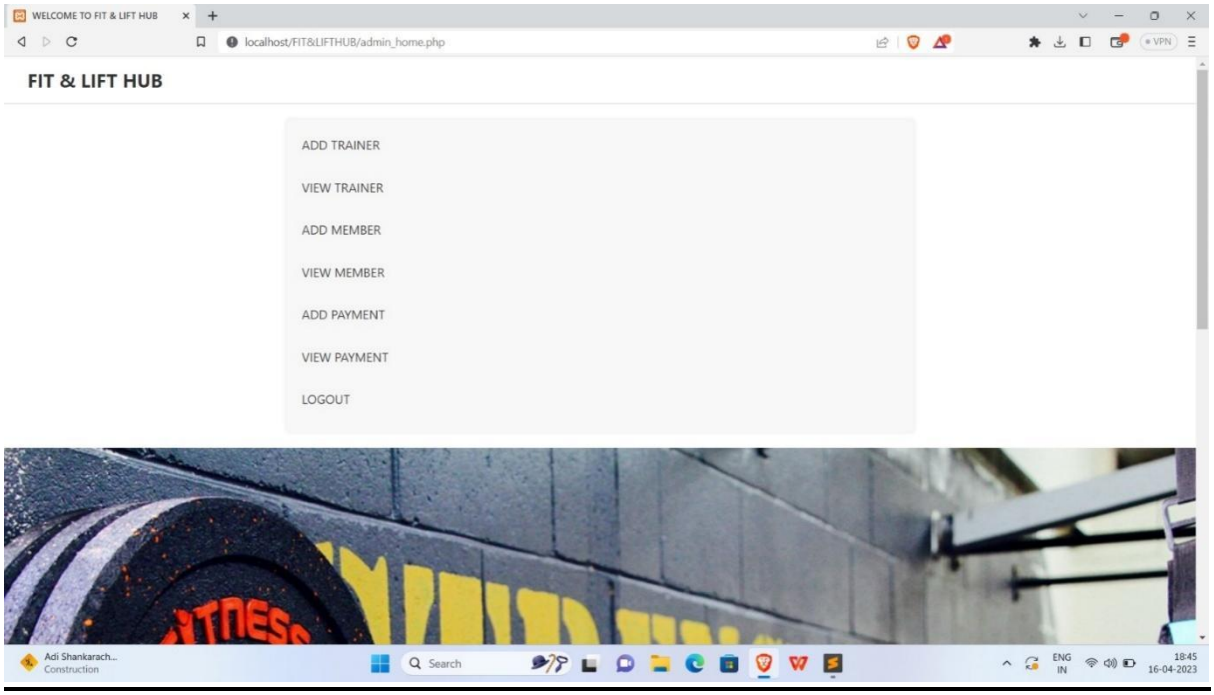
```
.footer {
  background-color: #333;
  color: white;
  text-align: center;
  padding: 10px;
  position: fixed;
  bottom: 0;
  width: 100%;
}
    </style>
</head>
<body class="back">
    <nav>
        <a class="title" href="index.php">FIT & LIFT HUB</a>
        <a href="trainer_login.php">TRAINER LOGIN</a>
        <a href="member_login.php">MEMBER LOGIN</a>
        <a href="aboutus.php">ABOUT</a>
        <a href="contactus.php">CONTACT US</a>
    </nav>
        <div class="login">
        <h1 class="heading">ADMIN LOGIN</h1>
        <div class="log">
            <?php

            if(isset($_POST["login"])){
                $sql = "SELECT * from admin where ANAME='{$_POST["aname"]}' and APASS='{$_POST["apass"]}'";
                $res = $db->query($sql);

                if($res->num_rows>0){
                    $ro = $res->fetch_assoc();
                    $_SESSION["AID"]=$ro["AID"];
                    $_SESSION["ANAME"]=$ro["ANAME"];
                    echo "<script>window.open('admin_home.php','_self');</script>";
                } else {
                    echo "<div class='error'>Invalid Username or password</div>";
                }
            }

            ?>

            <form method="POST" action="">
             <label>USERNAME</label>
             <input type="text" name="aname" required class="input">
             <label>PASSWORD</label>
             <input type="Password" name="apass" required class="input"><br>
             <button type="submit" class="btn" name="login">LOGIN</button>
             </form>
        </div>
    </div>
    <div class="footer">
        <footer><p>Copy Right &copy; FIT&LIFTHUB </p></footer>
    </div>
</body>
</html>
```



5.2 Admin Home

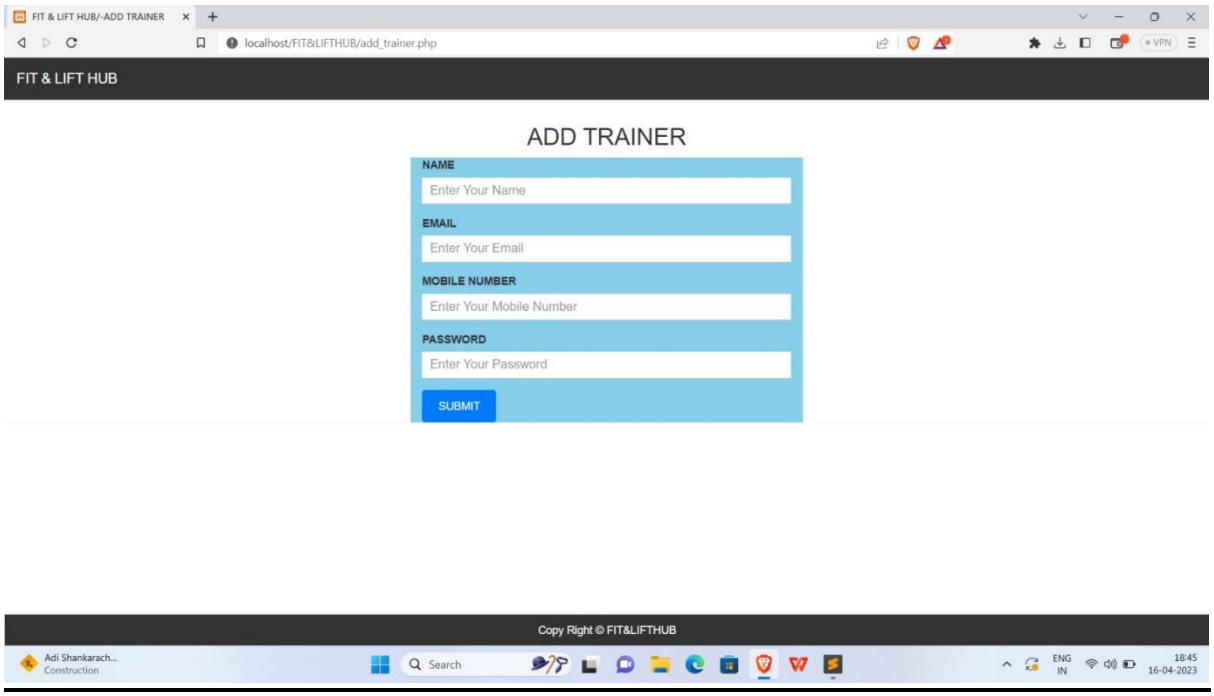
admin_home.php

```
admin_home.php
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>WELCOME TO FIT & LIFT HUB</title>
    <link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css"
        integrity="sha384-ggOyR0iXCbMQv3Xipma34MD+dH/1fQ784/j6cY/iJTQUOhcWr7x9JvoRxT2MZw1T"
        crossorigin="anonymous">
    <style>
        .panel {
            background-color: #f7f7f7;
            border-radius: 5px;
            box-shadow: 0 0 5px rgba(0, 0, 0, 0.1);
            padding: 10px;
            margin: 20px auto;
            max-width: 800px;
        }
        .panel ul {
            list-style: none;
            margin: 0;
            padding: 0;
        }
        .panel ul li {
            margin-bottom: 10px;
        }
        .panel ul li a {
            color: #333;
            display: block;
            padding: 10px;
            text-decoration: none;
            transition: all 0.3s ease;
            border-radius: 5px;
        }
        .panel ul li a:hover {
            background-color: #e6e6e6;
        }
        nav {
            background-color: #fff;
```

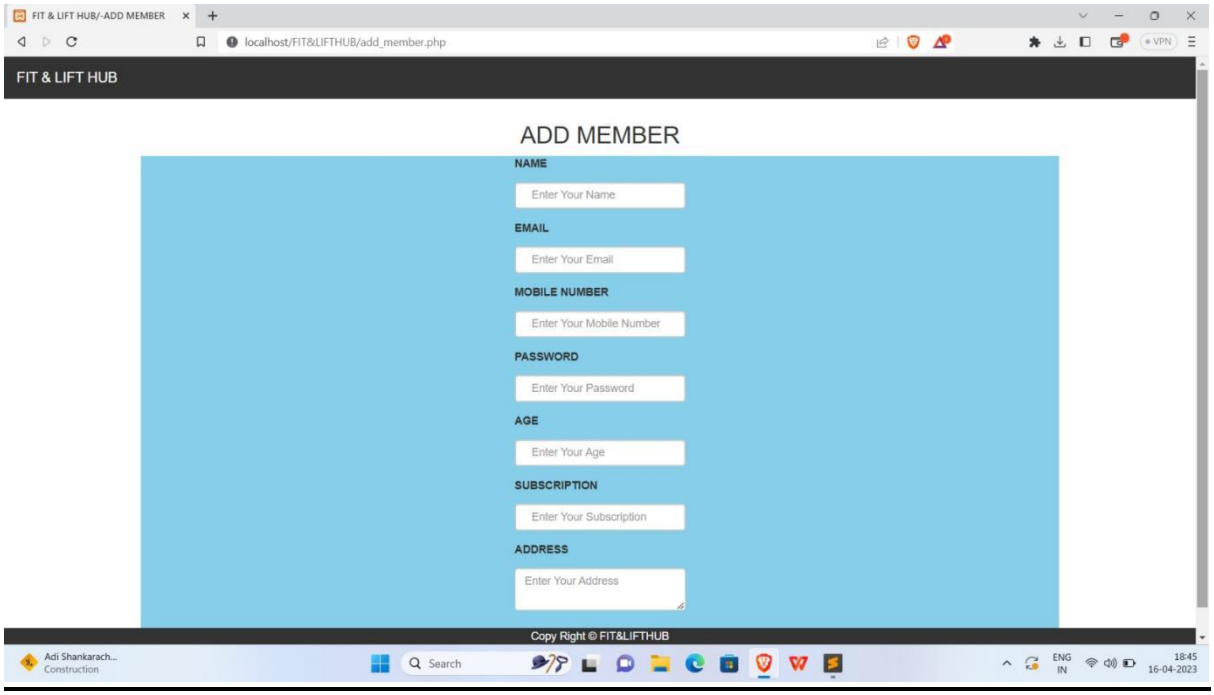
admin_home.php

```
40                  border-bottom: 1px solid #ccc;
41                  display: flex;
42                  align-items: center;
43                  height: 60px;
44                  padding: 0 20px;
45              }
46              nav a.title {
47                  color: #333;
48                  font-size: 24px;
49                  font-weight: bold;
50                  text-decoration: none;
51                  margin-left: 10px;
52              }
53              nav a.logo {
54                  display: inline-block;
55                  height: 40px;
56                  width: 40px;
57                  background-size: contain;
58                  background-repeat: no-repeat;
59                  margin-right: 10px;
60              }
61              .footer {
62                  background-color: #333;
63                  color: #fff;
64                  padding: 20px;
65                  text-align: center;
66                  margin-top: 50px;
67              }
68              .footer p {
69                  margin: 0;
70              }
71          </style>
72      </head>
73      <body>
74      <div>
75          <nav>
76              <a class="title" href="">FIT & LIFT HUB</a>
77          </nav>
78          <div class="panel">
79        <ul>
```

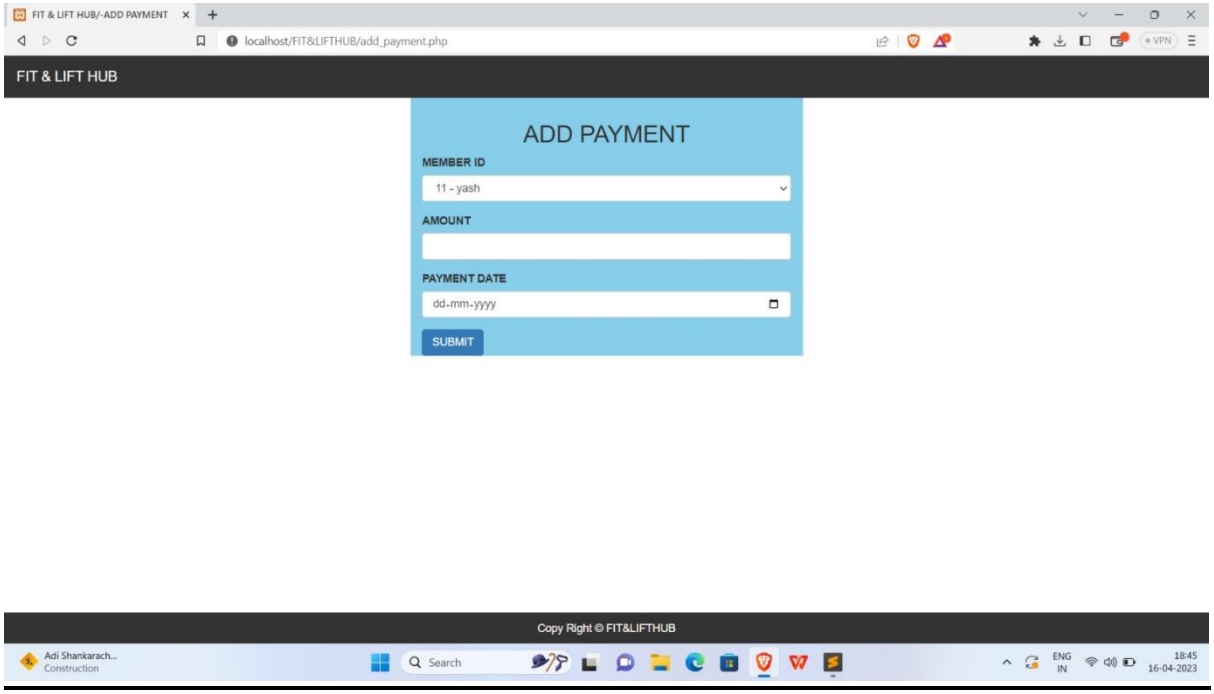
```html
      <li><a href="add_trainer.php">ADD TRAINER</a></li>
      <li><a href="display_trainer.php">VIEW TRAINER</a></li>
      <li><a href="add_member.php">ADD MEMBER</a></li>
      <li><a href="display_member.php">VIEW MEMBER</a></li>
      <li><a href="add_payment.php">ADD PAYMENT</a></li>
      <li><a href="display_payment.php">VIEW PAYMENT</a></li>
      <li><a href="admin_logout.php">LOGOUT</a></li>
    </ul>
  </div>
  </div>
  <img src="image/bg.png" width="100%">
      <div class="footer">
          <footer><p>Copy Right &copy; FIT&LIFTHUB </p></footer>
      </div>
  </body>
  </html>
```



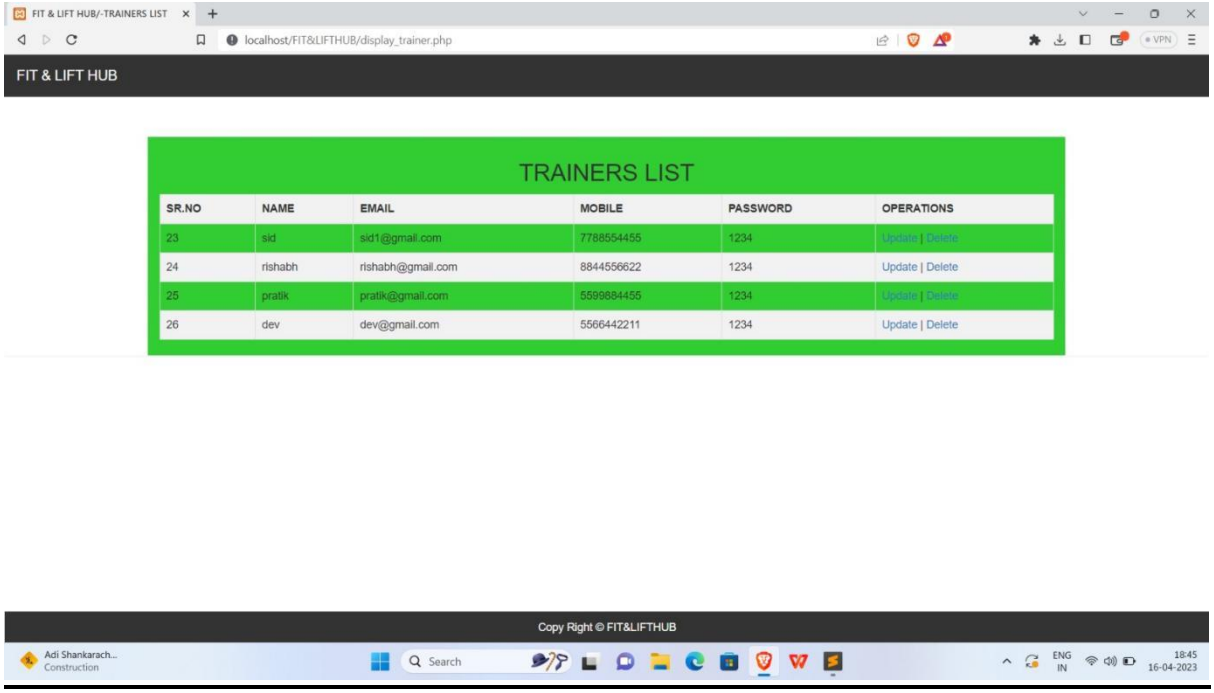
5.3 Add Trainer



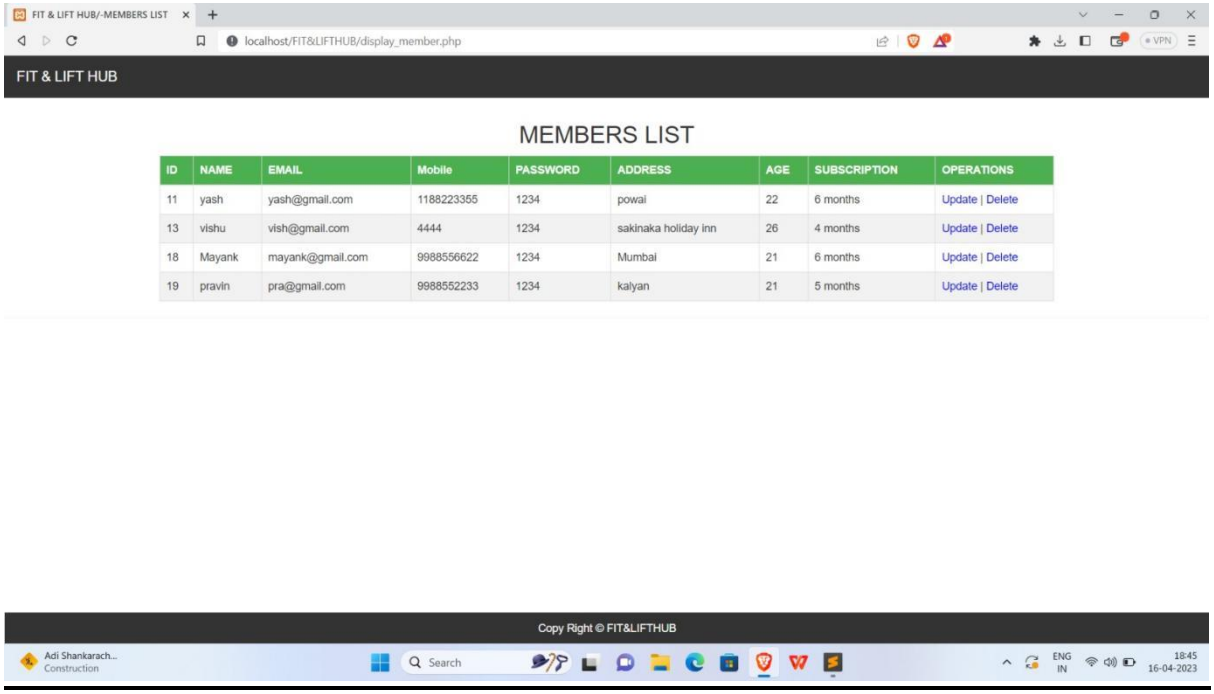
5.4 Add Member



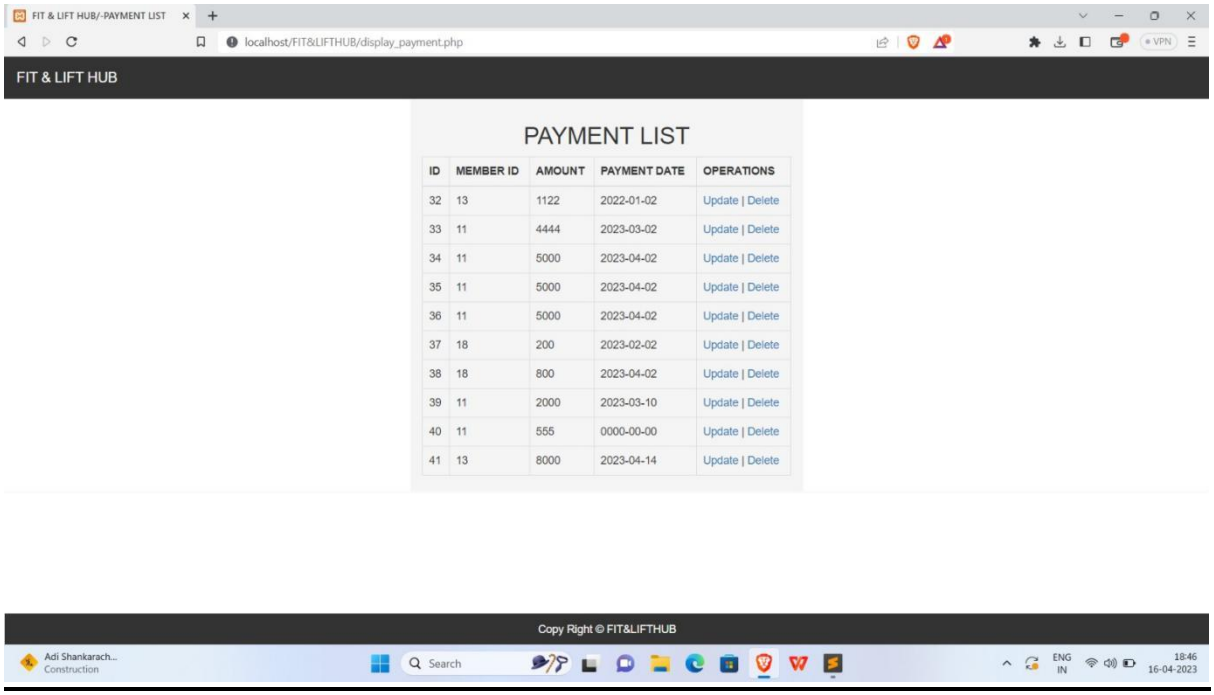
5.5 Add Payment



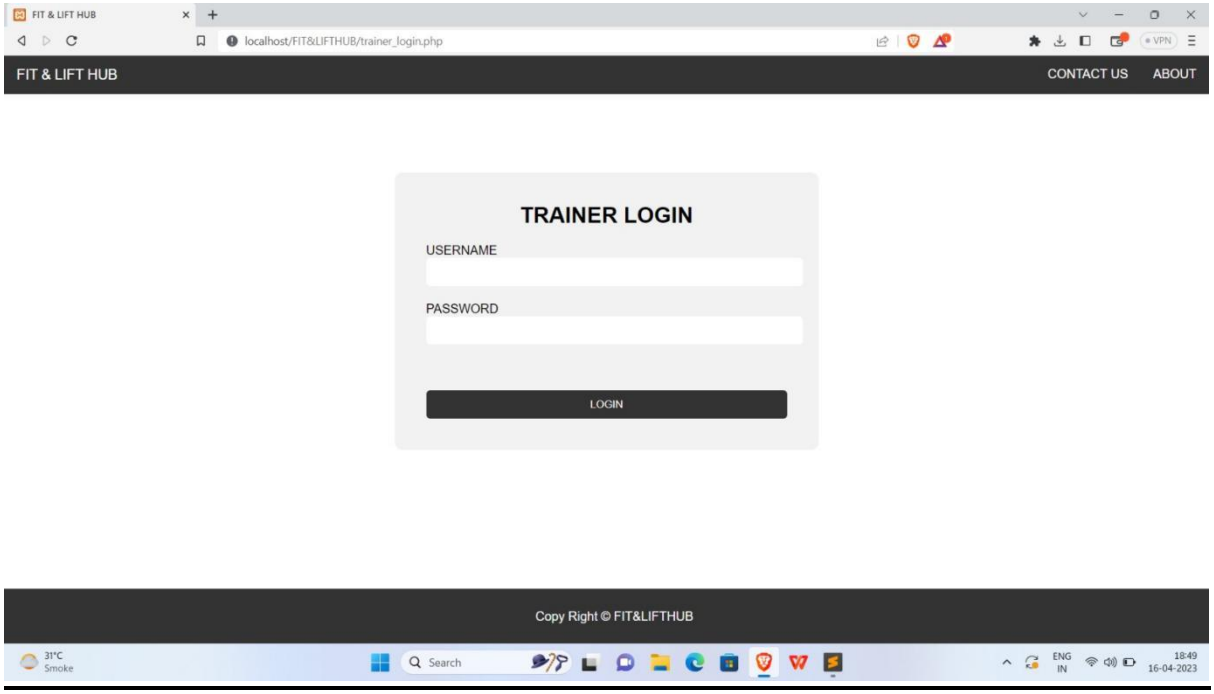
5.6 View Trainers



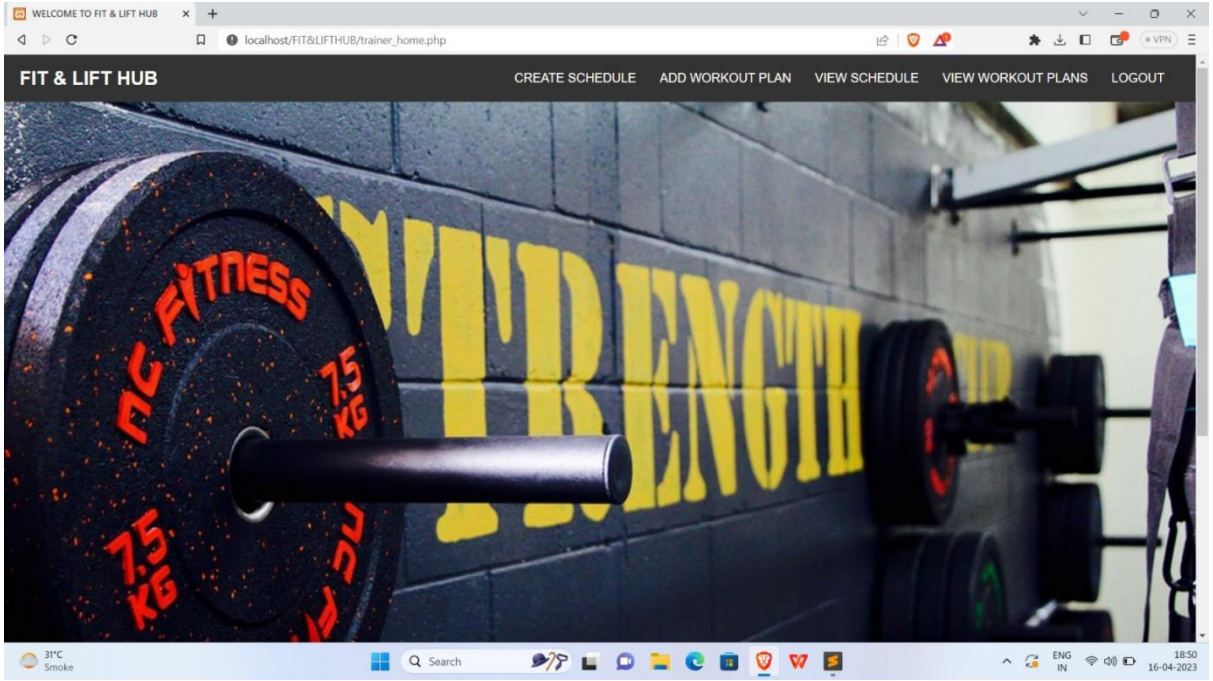
5.7 View Members



5.8 View Payment

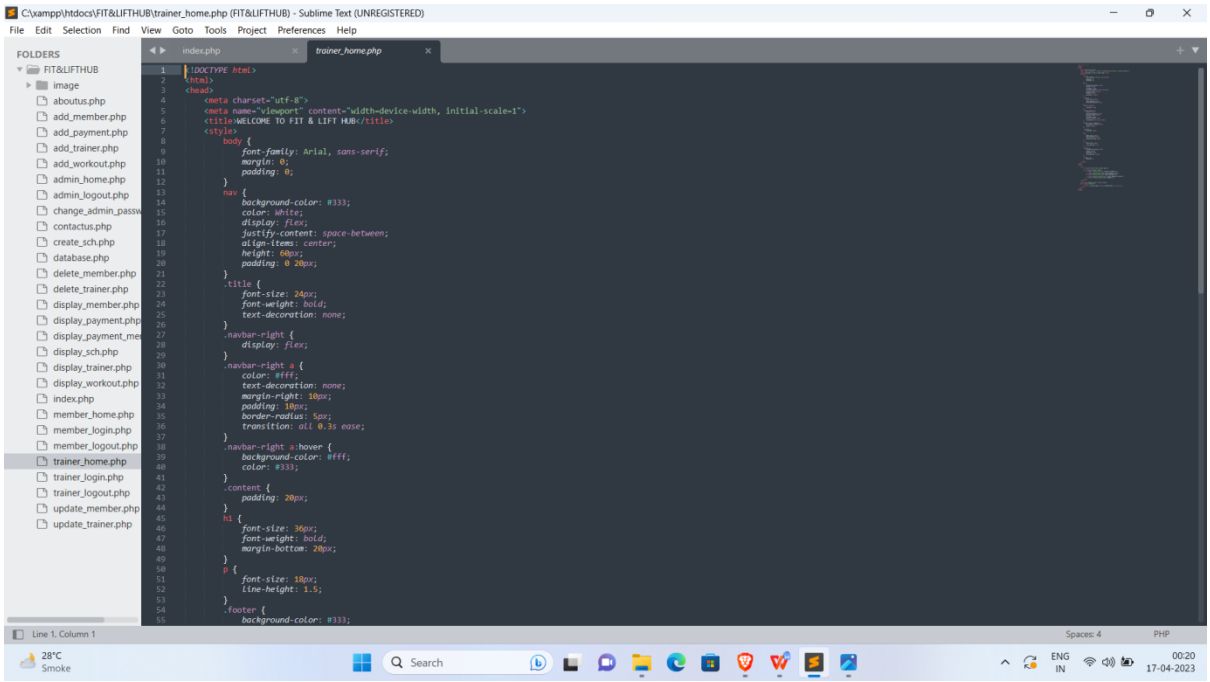


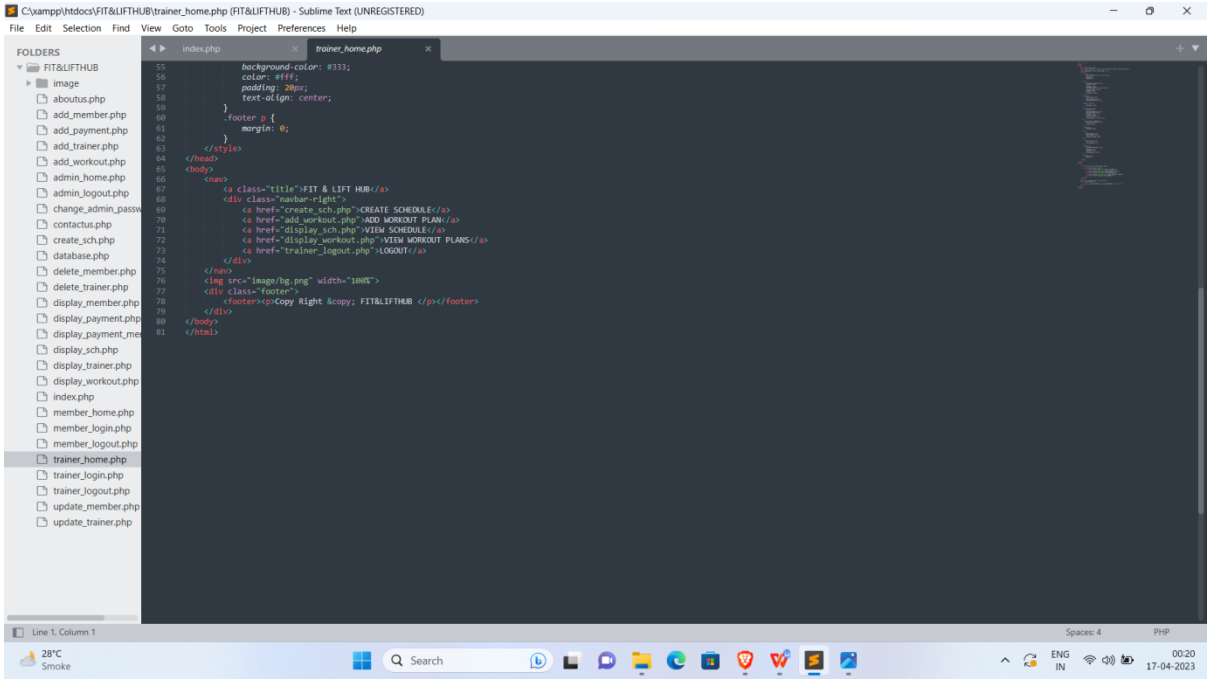
5.9 Trainer Login

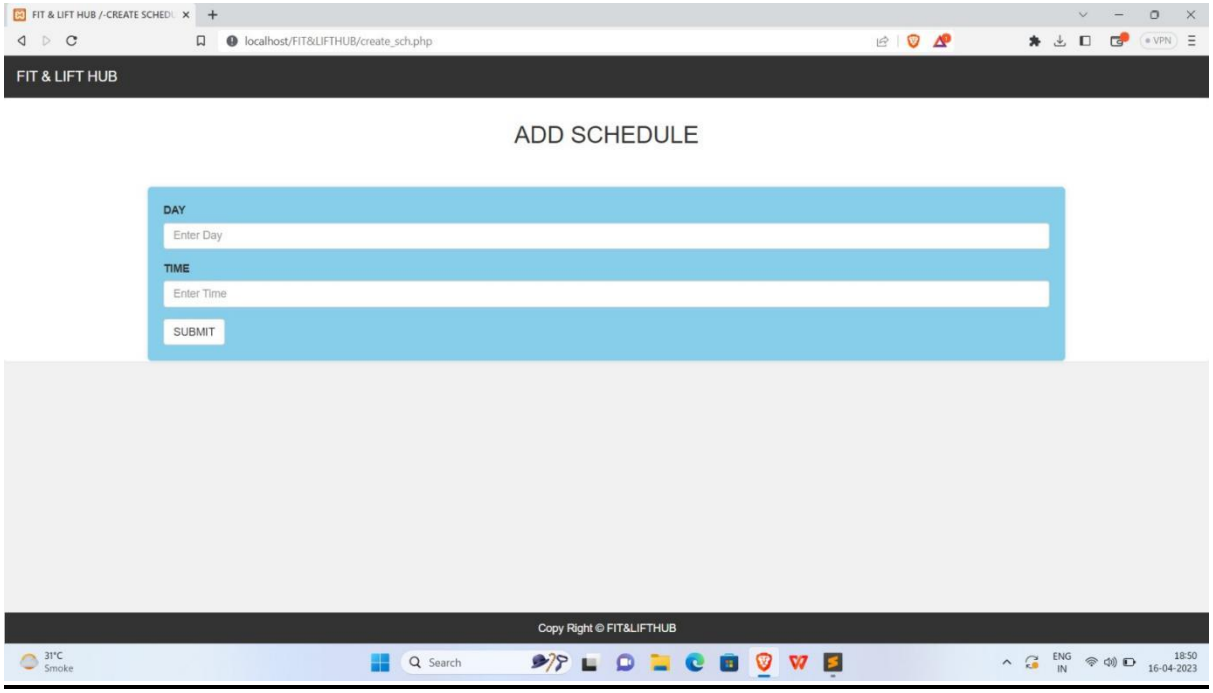


5.10 Trainer Home

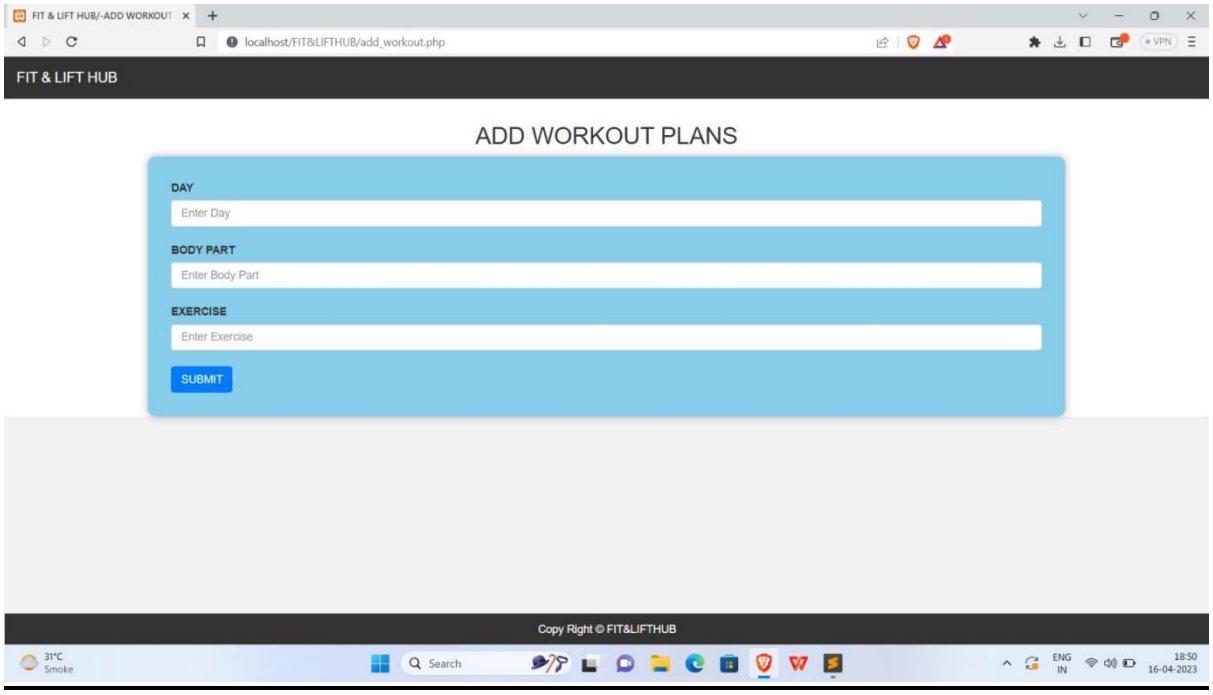
trainer_home.php



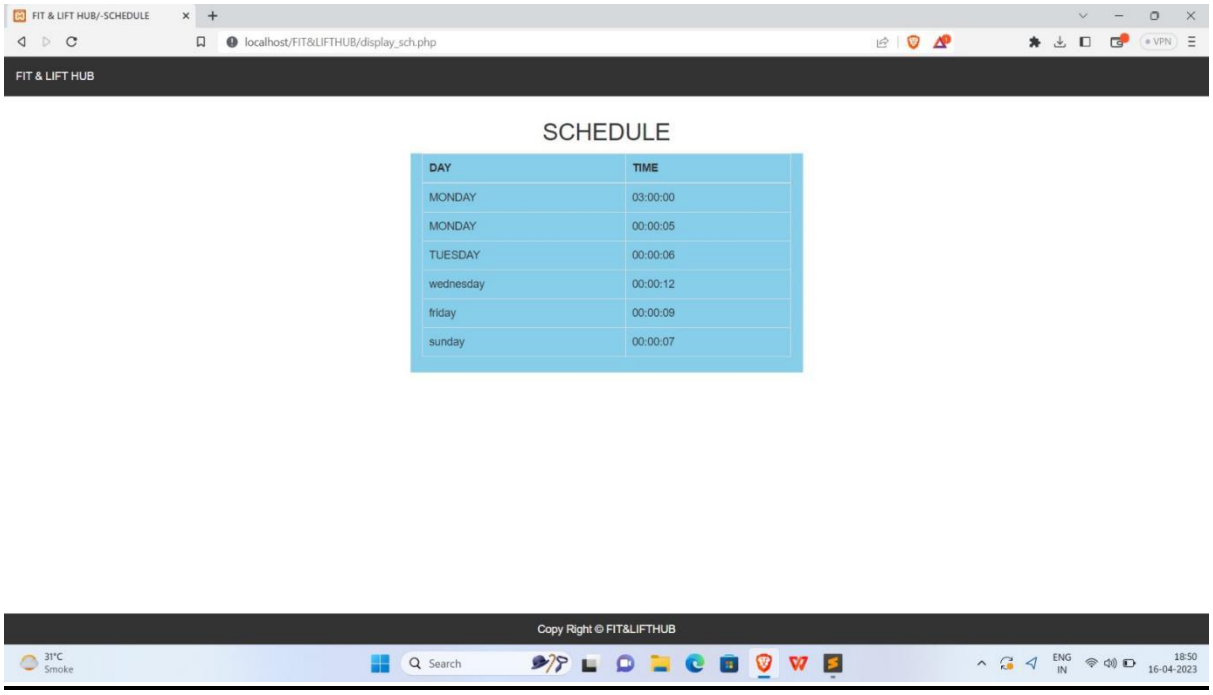




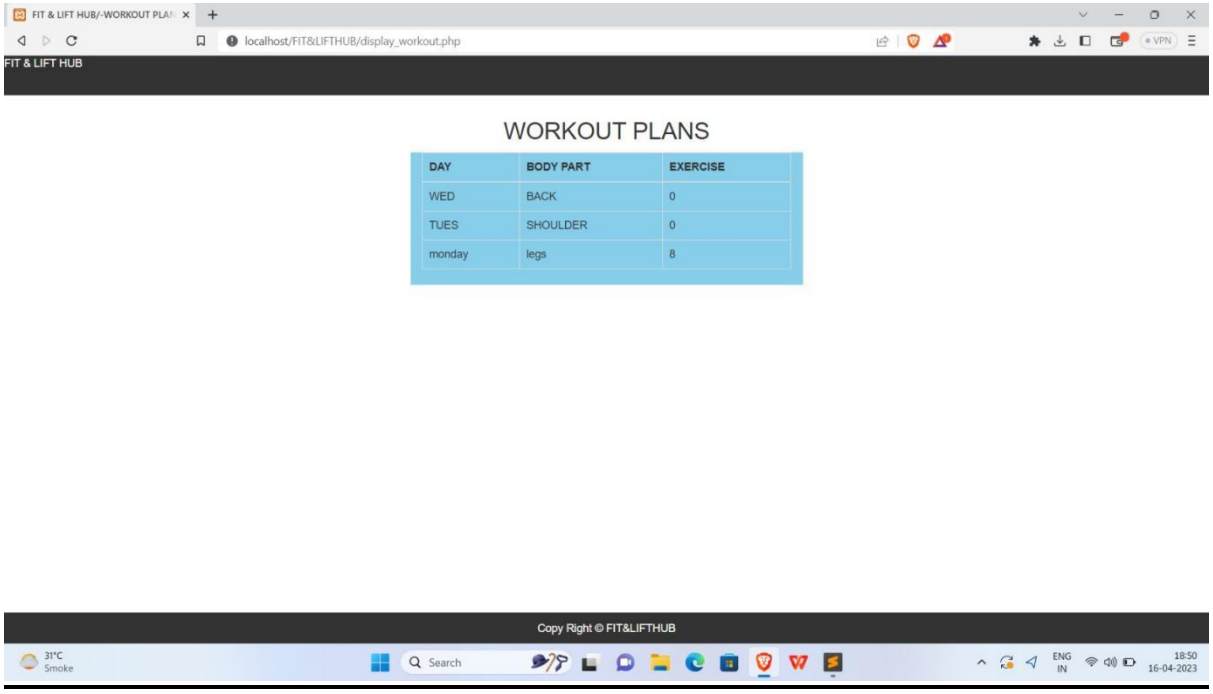
5.11 Add Schedule



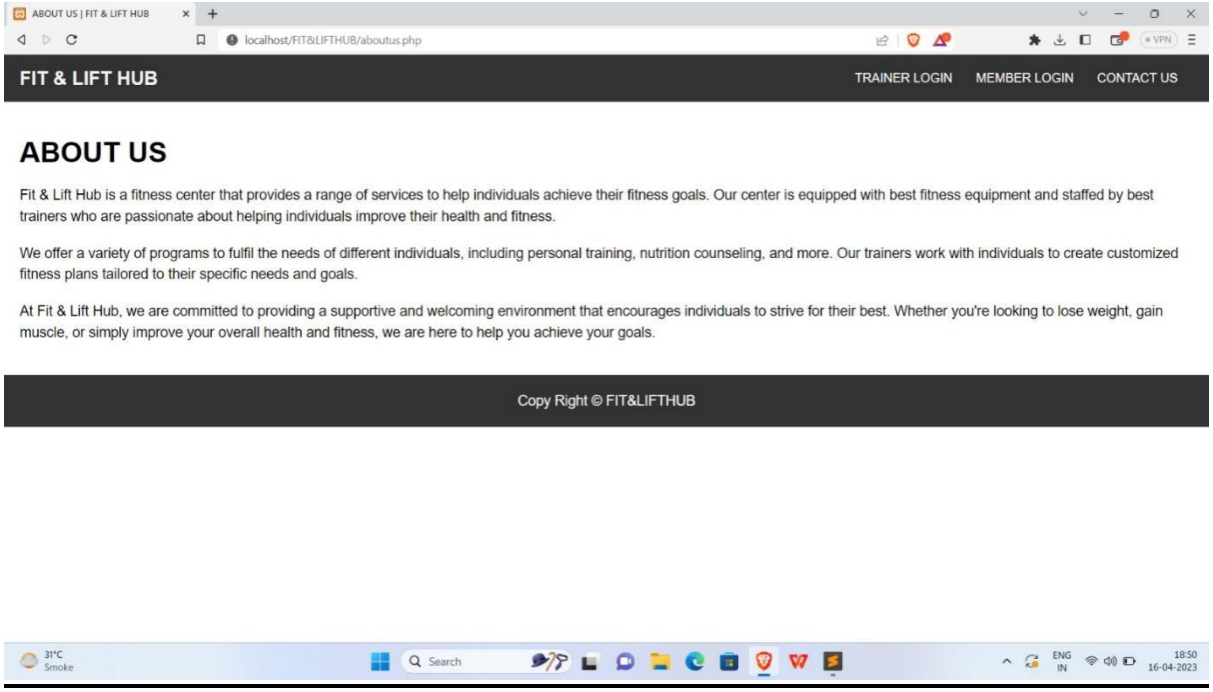
5.12 Add Workout Plans



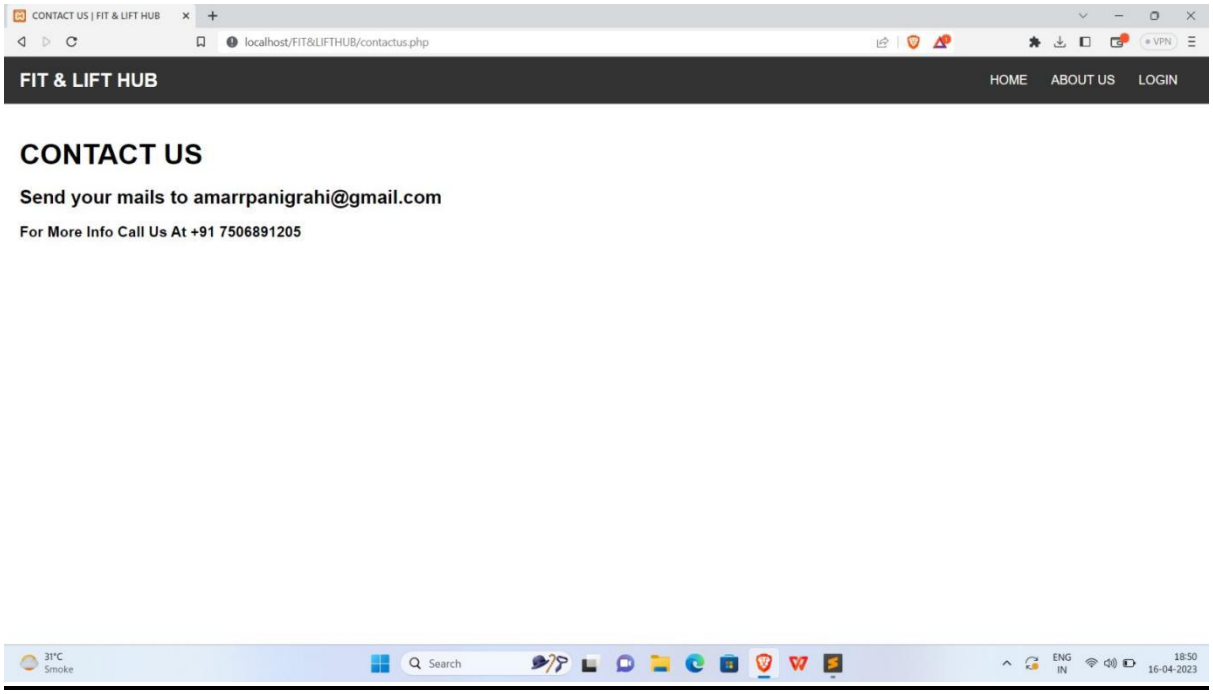
5.13 View Schedule



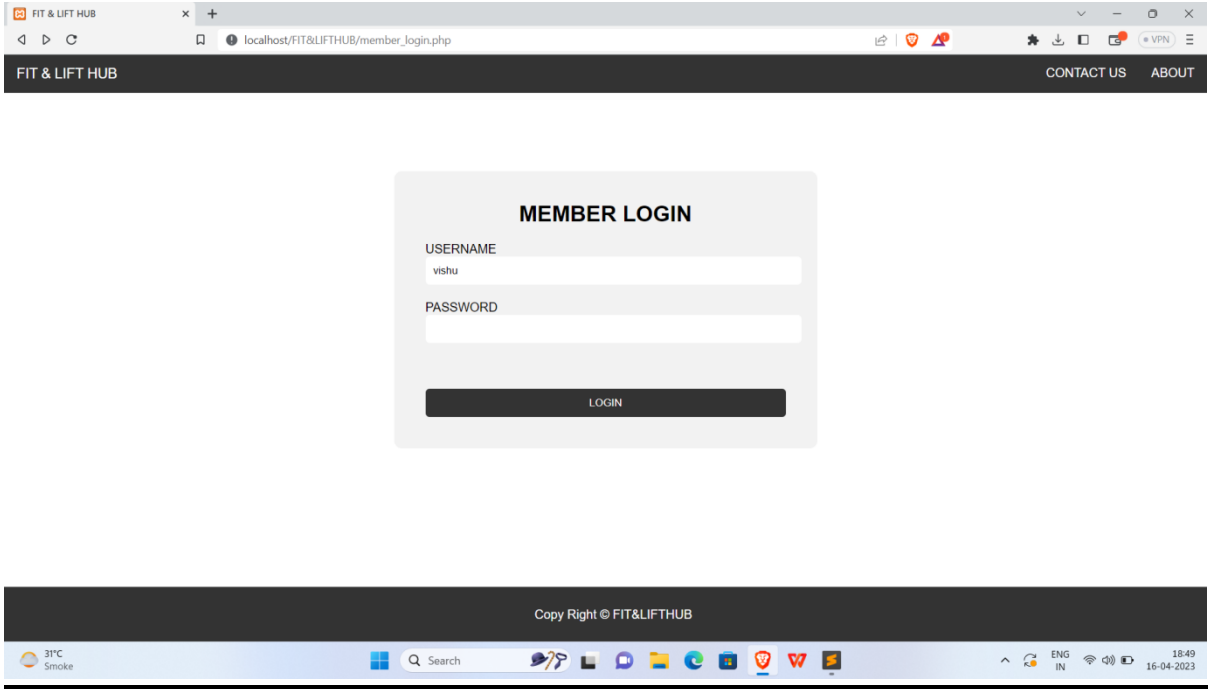
5.14 View Workout Plans



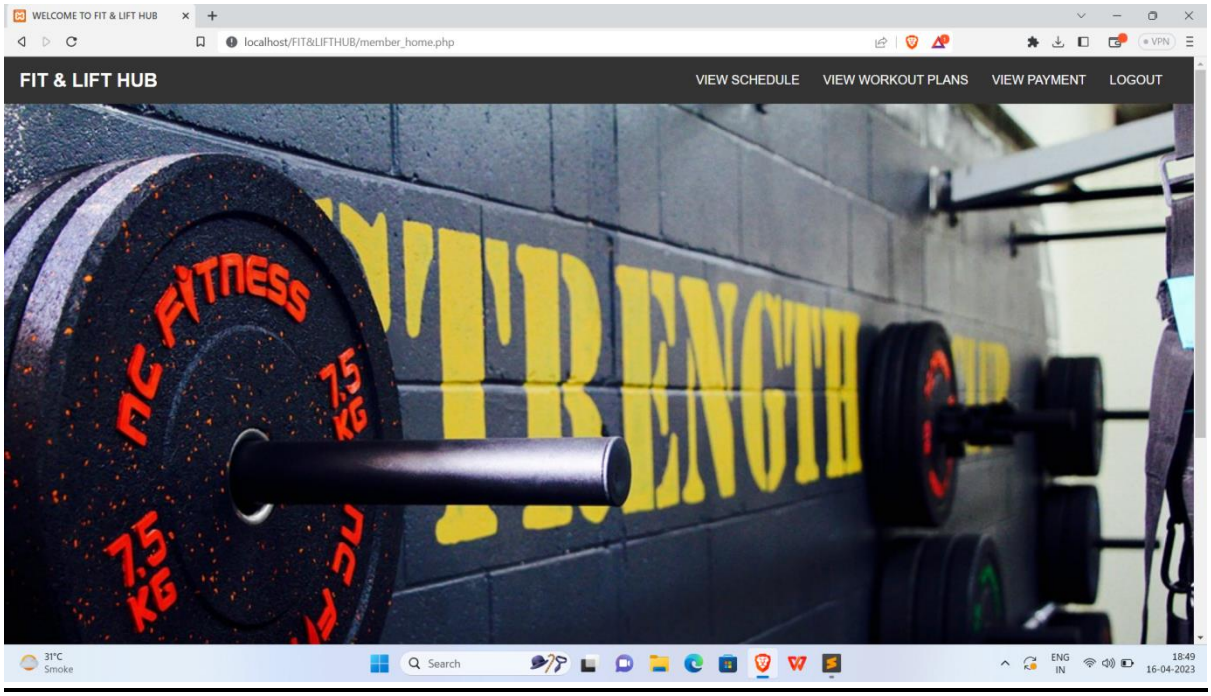
5.15 About



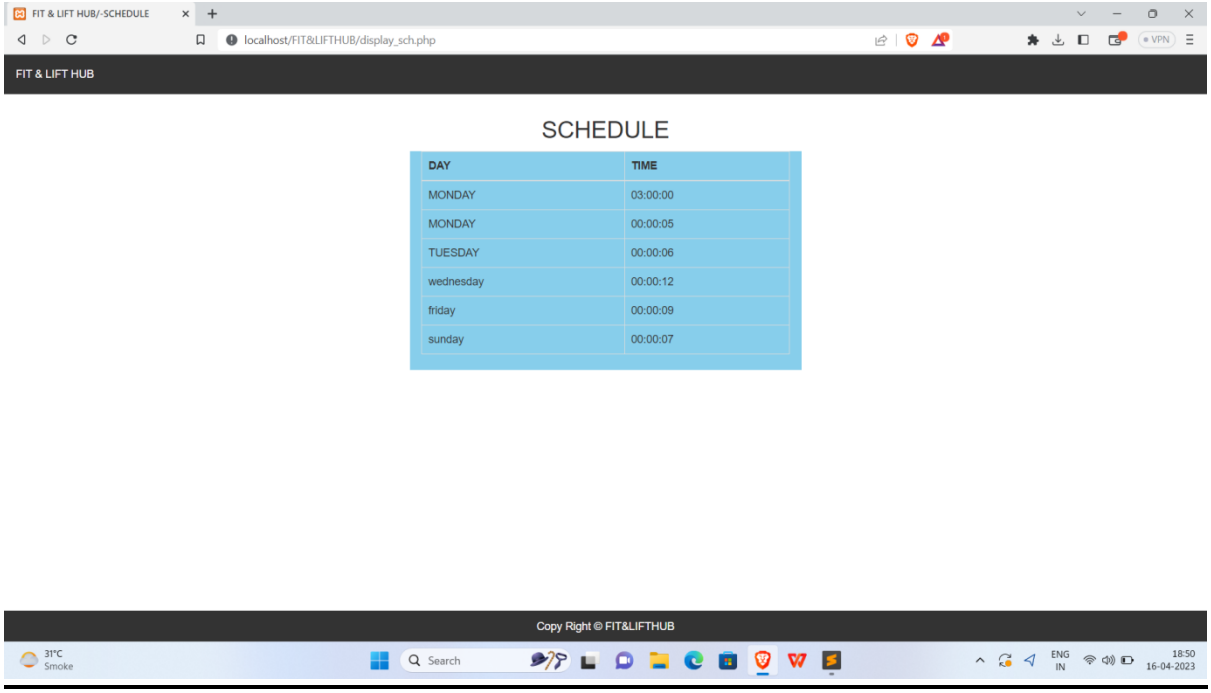
5.16 Contact Us



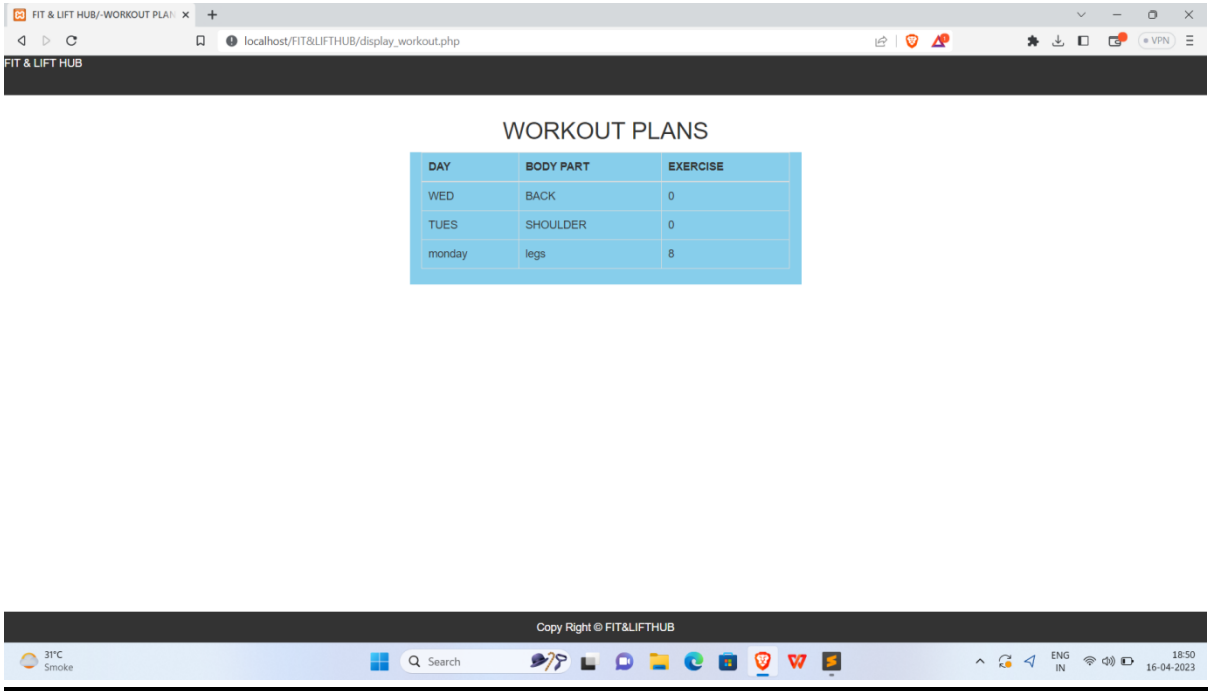
5.17 Members Login



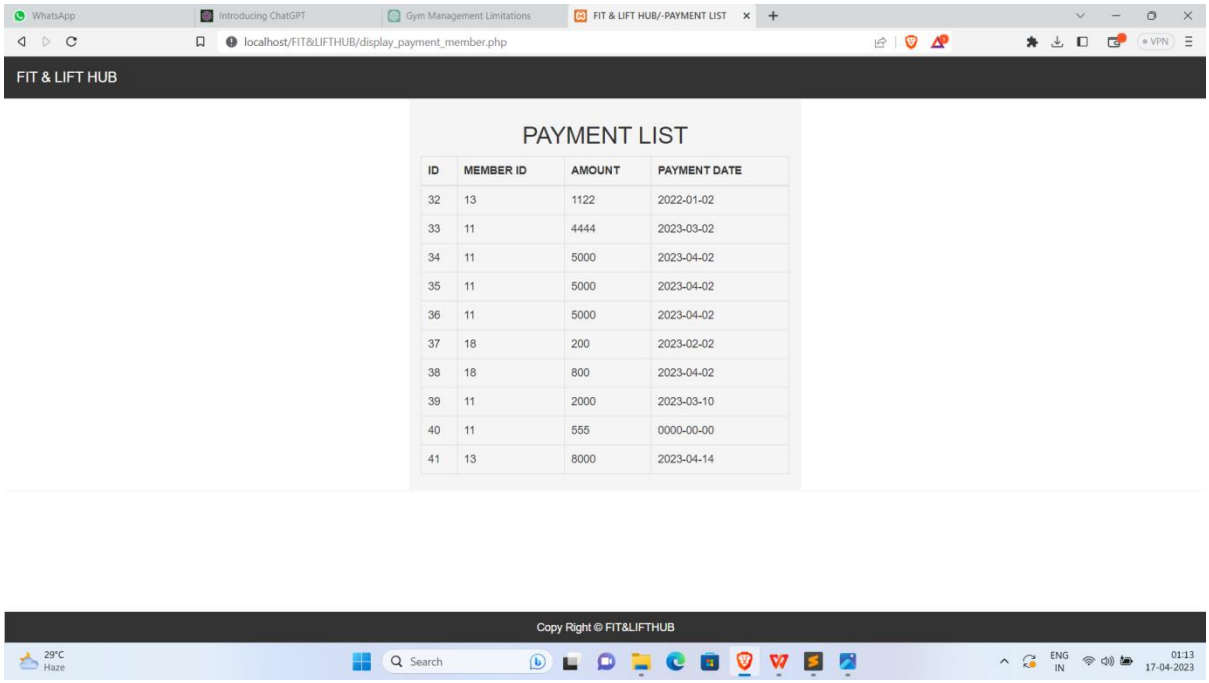
5.18 Member Home



5.19 View Schedule



5.20 View Workout Plans



ID	MEMBER ID	AMOUNT	PAYMENT DATE
32	13	1122	2022-01-02
33	11	4444	2023-03-02
34	11	5000	2023-04-02
35	11	5000	2023-04-02
36	11	5000	2023-04-02
37	18	200	2023-02-02
38	18	800	2023-04-02
39	11	2000	2023-03-10
40	11	555	0000-00-00
41	13	8000	2023-04-14

5.21 View Payment

CHAPTER 6

IMPLEMENTATION AND TESTING

6.1 Software Testing

Software testing is the process of evaluating the performance of software applications to find out if the software meets the requirements set out in and to identify defects to ensure that products are error-free in order to produce quality products. Test Procedures include the process of executing a program or application to detect software errors (viruses or other defects) and to verify appropriate software equipment. Software testing involves running a software component or physical device to measure one or multiple aspects of interest.

Since the number of possible tests is nearly infinite even for simple software products, all test software uses some strategy to select the tests possible with time and resources. Therefore, the software test usually (but is not limited to) attempts to complete a program or application to find software errors (bugs or other flaws). The test job is an iterative process because it can fix one bug while clarifying other deep bugs and even creating new ones. Software testing can provide users or sponsors with objective, independent information about software quality and the risk of bugs. Software evaluation is allowed as long as complete software (even if part of it has been completed) is available. The overall approach to software development usually dictates when and how tests are performed. For example, most testing in a staged process is performed after a system has been identified and used as the program under test.

Conversely, according to the Agile methodology, requirements, programming, and testing are usually done concurrently.

6.1.1 Overview:

While testing can determine the correctness of the software based on some assumptions on some assumptions, testing cannot identify all flaws in the software. Instead, it provides critique or comparison that compares the state and behaviour of things with empirical prophecies— principles or methods by which one can identify problems. These estimates may include, but are not limited to, specifications, contracts, product comparisons, previous products of the same product, assumptions about the target needs of brands, user or customer expectations, standards, laws or other standards. The main purpose of TEST is to analyse software bugs to find and fix bugs. The scope of software testing usually includes reviewing the code and its execution in different environments and situations, and examining aspects of the code : how it does what it's supposed to do and what it takes to do it. In today's software development culture, the test organization is separate from the development team. Teams with different roles were tested. Information obtained from software testing is used to improve the software development process.

6.1.2 Defects and failures

Not all software defects are caused by coding errors. One common source of defects is requirement gaps e.g., unrecognized requirements that result in errors of omission by the program designer. Requirement gaps can often be non-functional requirements such as scalability, testability, maintainability, usability, performance, and security. Software faults occur through the following processes. A programmer makes an error (mistake), which results in a defect (fault, bug) in the software source code. In the event that this imperfection is executed, in specific circumstances the framework will deliver wrong outcomes, causing a disappointment. Not all deformities will essentially bring about disappointments. An imperfection can transform into a disappointment when the climate is changed. Instances of these progressions in climate incorporate the product being run on another PC equipment stage, changes in source information, or associating with various programming. A solitary deformity might bring about an extensive variety of disappointing side effects.

6.1.3 Input combinations and preconditions

A fundamental problem with software testing is that testing under all combinations of inputs and preconditions (initial state) is not feasible, even with a simple product. This means that the number of defects in a software product can be very large and defects that occur infrequently are difficult to find in testing. Exceptionally emotional; something that is adequate worth to one individual might be insufferable to another. Programming engineers can't test everything, except they can utilize a combinatorial test plan to distinguish the base number of tests expected to get the inclusion they need. Combinatorial test configuration empowers clients to get more prominent test inclusion with less tests. Whether they are searching for speed or test profundity, they can utilize combinatorial test plan strategies to incorporate organized variety into their experiments.

6.2 Testing Approach

6.2.1 Static vs. dynamic testing

There are many approaches available in software testing. Surveys, walkthroughs, or investigations are alluded to as static testing, while executing modified code with a given arrangement of experiments is alluded to as unique testing.Static testing is many times verifiable, such as editing, in addition to while programming apparatuses/word processors check source code design or compilers (pre-compilers) check grammar and information stream as static program examination. Dynamic testing happens when the actual program is run. Dynamic testing might start before the program is 100 percent complete to test specific segments of code and are applied to discrete capabilities or modules.Typical techniques for these are either using stubs/drivers or execution from a debugger environment. Static testing involves verification, whereas dynamic testing also involves validation. Verification is the process, to ensure that whether we are building the product right i.e., to verify the requirements which we have and to verify whether we are developing the product accordingly or not. Activities involved here are Inspections, Reviews and Walkthroughs. Validation is the process, whether we are building the right product i.e., to validate the product which we have developed is right or not.

6.2.2 Exploratory approach:

Exploratory testing is a way to deal with programming testing that is succinctly portrayed as concurrent learning, test plan and test execution. Cem Kaner, who begat the term in 1984, characterizes exploratory testing as "a style of programming testing that underscores the individual flexibility and obligation of the singular analyzer to constantly enhance the nature of his/her work by treating test-related learning, test configuration, test execution, and experimental outcome translation as commonly strong exercises that disagreement lined up all through the undertaking.

" 6.2.3 The "box" approach:

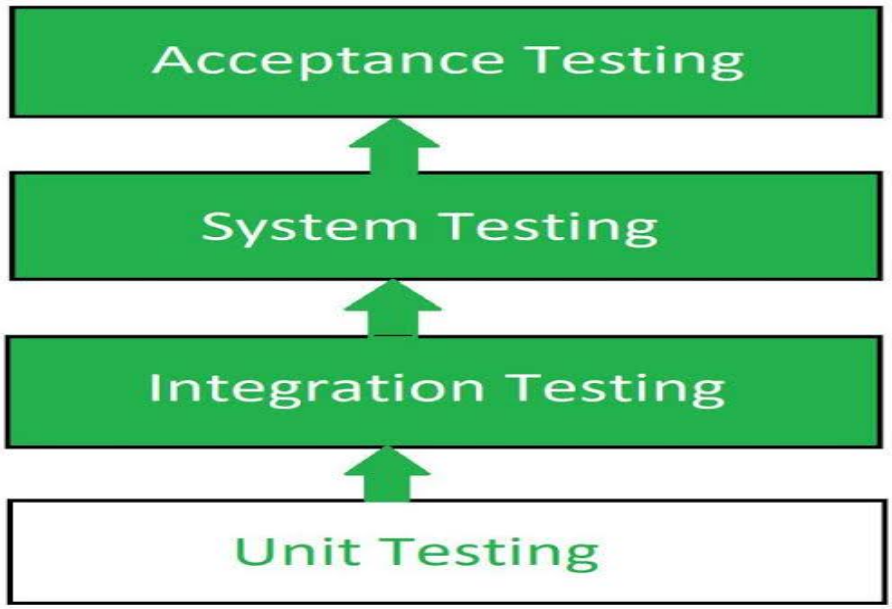
Software testing methods are traditionally divided into white-box testing and black-box testing. These two methodologies are utilized to portray the perspective that the analyzer takes while planning experiments. A mixture approach called dim box testing may likewise be applied to programming testing technique. With the concept of grey-box testing—which develops tests from specific design elements—gaining prominence, this "arbitrary distinction" between black- and white-box testing has faded somewhat.

● **White-box testing:**

White-box testing (also known as clear box testing, glass box testing, transparent box testing, and structural testing) verifies the internal structures or workings of a program, as opposed to the functionality exposed to the end-user. In white-box testing, an interior point of view of the framework (the source code), as well as programming abilities, are utilised to configuration experiments. The analyzer picks contributions to practise ways through the code and decide the suitable results. This is comparable to testing hubs in a circuit, e.g., in-circuit testing (ICT). While white-box testing can be applied at the unit, integration, and system levels of the software testing process, it is usually done at the unit level.It can test ways inside a unit, ways between units during joining, and between subsystems during a framework level test. However this technique for test configuration can uncover numerous blunders or issues, it probably won't recognize unimplemented parts of the detail or missing prerequisites.

6.3 Testing Levels

There are something like three degrees of testing: unit testing, Integration testing, and System testing. Notwithstanding, a fourth level, acknowledgment testing, might be incorporated by engineers. This might be as functional acknowledgment testing or be basic end-client (beta) testing, testing to guarantee the product measures up to utilitarian assumptions. Tests are regularly gathered into one of these levels by where they are included in the product advancement process, or by the degree of particularity of the test.



6.3.1 Unit testing:

Unit testing refers to tests that verify the functionality of a specific section of code, usually at the function level. In an object-oriented environment, this is usually at the class level, and the minimal unit tests include the constructors and destructors. These sorts of tests are generally composed by engineers as they work on code to guarantee that the particular capability is filling in true to form. One capability could have numerous tests, to get corner cases or different branches in the code. Unit testing alone can't check the usefulness of a piece of programming, yet rather is utilised to guarantee that the structure blocks of the product work freely from one another. Unit testing is a product improvement process that includes a synchronised utilisation of an expansive range of deformity counteraction and identification methodologies to decrease programming advancement dangers, time, and expenses.It is performed by the product designer or architect during the development period of the product advancement lifecycle. Unit testing plans to dispense with development blunders before code is elevated to extra testing; this system is expected to expand the nature of the subsequent programming as well as the productivity of the general advancement process. Depending on the organisation's expectations for software development, unit testing might include static code analysis, data-flow analysis, metrics analysis, peer code reviews, code coverage analysis and other software testing practices.

6.3.2 Integration testing:

Integration testing is any type of software testing that seeks to verify the interfaces between components against a software design. Programming parts might be coordinated in an iterative manner or all together Typically the previous is viewed as a superior practice since it permits interface issues to be found all the more rapidly and fixed. Mix testing attempts to uncover deserts in the connection points and communication between coordinated parts (modules). Dynamically bigger gatherings of tried programming parts compared to components of the structural plan are coordinated and tried until the product functions as a framework.

6.3.3 System testing:

System testing tests a completely integrated system to verify that the system meets its requirements. For example, a system test might involve testing a login interface, then creating and editing an entry, plus sending or printing results, followed by summary processing or deletion (or archiving) of entries, then logoff.

6.3.4 Operational Acceptance Testing:

Operational acceptance is used to conduct operational readiness (prerelease) of a product, service or system as part of a quality management system. OAT is a typical kind of non-practical programming testing, utilized essentially in programming improvement and programming upkeep projects. This sort of testing centers around the functional availability of the framework to be upheld, or to turn out to be important for the creation climate. Consequently, it is otherwise called functional preparation testing (ORT) or Operations readiness and assurance (OR&A) testing. Functional testing within OAT is limited to those tests that Operations readiness and assurance are required to verify the non-functional aspects of the system. In addition, the software testing should ensure that the portability of the system, as well as working as expected, does not also damage or partially corrupt its operating environment or cause other processes within that environment to become inoperative.

6.4 Testing Types, Techniques and Tactics:

Different labels and ways of grouping testing may be testing types, software testing tactics or techniques.

6.4.1 Installation Testing: Most software systems have installation procedures that are needed before they can be used for their main purpose. Testing these procedures to achieve an installed software system that may be used is known as installation testing.

6.4.2 Acceptance Testing:

Acceptance testing can mean one of two things:

● A smoke test is used as a build acceptance test prior to further testing, e.g., before integration or regression.

● Acceptance testing performed by the client, frequently in their lab climate on their own equipment, is known as client acknowledgment testing (UAT). Acceptance testing might proceed as a component of the handoff cycle between any two periods of improvement.

6.4.3 Alpha Testing

Alpha testing is simulated or actual operational testing by potential users/customers or an independent test team at the developers' site. Alpha testing is often employed for off-theshelf software as a form of internal acceptance testing before the software goes to beta testing.

6.4.4 Beta Testing:

Beta testing comes after alpha testing and can be considered a form of external user acceptance testing. Variants of the product, known as beta forms, are delivered to a restricted crowd beyond the programming group known as beta testers. The product is delivered to gatherings so that further testing can guarantee the item has not many blames or bugs. Beta variants can be made accessible to the open public to build the criticism field to a maximal number of future clients and to convey esteem prior, for a drawn out or even endless timeframe (perpetual beta).

6.4.5 Functional vs Non-Functional Testing:

Functional testing alludes to exercises that check a particular activity or capability of the code. These are normally tracked down in the code prerequisites documentation, albeit some improvement strategies work from use cases or client stories. Practical tests will generally respond to the subject of "might the client at any point do this" or "does this specific element work." Non-functional testing refers to aspects of the software that may not be related to a specific function or user action, such as scalability or other performance, behaviour under certain constraints, or security. Testing will determine the breaking point, the point at which extremes of scalability or performance leads to unstable execution. Non-functional requirements tend to be those that reflect the quality of the product, particularly in the context of the suitability perspective of its users.

6.4.6 Security Testing:

Security testing is essential for software that processes confidential data to prevent system intrusion by hackers. The International Organization for Standardization (ISO) defines this as a "type of testing conducted to evaluate the degree to which a test item, and associated data and information, are protected so that unauthorized persons or systems cannot use, read or modify them, and authorized persons or systems are not denied access to them."

6.4.7 Development Testing:

Development Testing is a product improvement process that includes the synchronized use of an expansive range of deformity counteraction and location systems to decrease programming improvement dangers, time, and expenses. It is performed by the product designer or specialist during the development period of the product advancement lifecycle. Advancement Testing expects to take out development mistakes before code is elevated to other testing; this technique is planned to expand the nature of the subsequent programming as well as the effectiveness of the general improvement process.

Depending upon the association's assumptions for programming improvement, Advancement Testing could incorporate static code examination, information stream investigation, measurements investigation, peer code surveys, unit testing, code inclusion examination, discernibility, and other programming testing rehearses.

6.5 Automation Testing:

Many programming groups are relying more and more on automated testing, especially groups that use test-driven development. There are numerous structures to compose tests in, and nonstop combination programming will run tests naturally every time code is looked into a variant control framework. While mechanization can't imitate all that a human can do (and every one of the manners in which they consider making it happen), it tends to be extremely valuable for relapse testing. Be that as it may, it requires an advanced test set-up of testing scripts to be really valuable.

6.5.1 Testing tools:

Program testing and fault detection can be aided significantly by testing tools and debuggers. Testing/debug tools include features such as:

● Program screens, allowing full or halfway observing of program code including:

● Guidance set test system, allowing total guidance level checking and follow offices.

● Hypervisor, permitting complete control of the execution of program code including:- Program animation, permitting step-by-step execution and conditional breakpoint at source level or in machine code.

● Code coverage reports.

● Formatted dump or symbolic debugging, tools allowing inspection of program variables on error or at chosen points.

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● Benchmarks, permitting run-time execution correlations with be made.

● Performance analysis (or profiling tools) that can help to highlight hot spots and resource usage. Some of these features may be incorporated into a single composite tool or an Integrated Development Environment (IDE)

6.5.2 Testing Artifacts

A software testing process can produce several artifacts. The actual artifacts produced are a factor of the software development model used, stakeholder and organizational needs.

- **Test plan:**

  A test plan is a document detailing the approach that will be taken for intended test activities. The plan may include aspects such as objectives, scope, processes and procedures, personnel requirements, and contingency plans. The test plan could come in the form of a single plan that includes all test types (like an acceptance or system test plan) and planning considerations, or it may be issued as a master test plan that provides an overview of more than one detailed test plan (a plan of a plan). A test plan can be, in some cases, part of a wide "test strategy" which documents overall testing approaches, which may itself be a master test plan or even a separate artifact.

- **Traceability matrix:**

  A traceability matrix is a table that correlates requirements or design documents to test documents. It is used to change tests when related source documents are changed, to select test cases for execution when planning for regression tests by considering requirement coverage.

- **Test case:**

An Test case ordinarily comprises a novel identifier, necessity references from a plan detail, preconditions, occasions, a progression of steps (otherwise called activities) to follow, input, yield, anticipated outcome, and the genuine outcome. Clinically characterized, an experiment is an information and a normal outcome. This can be essentially as pithy with respect to 'condition x your determined outcome is y', albeit ordinarily experiments portray in additional detail the info situation and what results may be normal. It can occasionally be a series of steps (but often steps are contained in a separate test procedure that can be exercised against multiple test cases, as a matter of economy) but with one expected result or expected outcome. The discretionary fields are an experiment ID, test step, or request of execution number, related requirement(s), profundity, test classification, creator, and check boxes for whether the test is automatable and has been robotized. Bigger experiments may likewise contain essential states or steps, and portrayals. A test case should also contain a place for the actual result.

- **Test script:**

A test script is a procedure or programming code that replicates user actions. Initially, the term was derived from the product of work created by automated regression test tools. A test case will be a baseline to create test scripts using a tool or a program.

- **Test suite:**

The most common term for a collection of test cases is a test suite. The test suite frequently likewise contains more itemized guidelines or objectives for every assortment of experiments. It most certainly contains a segment where the analyzer distinguishes the framework setup utilized during testing. A gathering of experiments may likewise contain essential states or steps, and portrayals of the accompanying tests.

CHAPTER 7

CONCLUSION & FUTURE SCOPE

7.1 Conclusion

While developing this project we have learnt a lot about HTML/CSS/JS/PHP/MySQL and working with database management, we have also learnt how to make the website user-friendly (easy to use and handle) by hiding the complicated parts of it from the users.

During the development process, we studied carefully and understood the criteria for making a software more demanding, we also realized the importance of maintaining a minimal margin for errors.

7.2 Limitations

System can be expensive, especially for small gyms with limited budgets. The cost includes hardware, software, training, and ongoing maintenance.

Have a pre-defined set of features and functionalities. It may not be possible to customize the system to fit the specific needs of every gym.

Gym staff may need to undergo training to learn how to use the system effectively. This can be time-consuming and may require additional resources.

Security risks: Gym management systems store sensitive information about members, such as their personal details and payment information. There is a risk of data breaches and cyber attacks, which can compromise the privacy and security of members.

Accessibility issues: Members who are not tech-savvy or do not have access to digital devices may face challenges in using the gym management system, which can result in dissatisfaction and frustration.

7.3 Future Scope

The future scope of Fit & Lift Hub is vast and promising. With advancements in technology, systems are expected to become more sophisticated, user-friendly, and efficient.

System could be integrated with wearable devices such as smartwatches and fitness trackers.

This would allow gym members to track their workouts, monitor their progress, and receive personalized feedback.

Artificial intelligence (AI) to offer personalized workout plans based on individual fitness levels, goals, and preferences.

AI could also be used to analyze member data and provide insights on gym usage, member behavior, and membership retention.

Overall, the future scope of System is exciting and has the potential to transform the fitness industry by offering innovative solutions that enhance member experience and drive business growth.

CHAPTER 8

REFERENCES

8.1 References

https://openai.com/blog/chatgpt

https://www.w3schools.com/

https://m.youtube.com/

https://stackoverflow.com/

www.academia.edu/

www.tutorialpoint.com/

8.2 Bibliography

Title of the project: FIT & LIFT HUB

Project Type: Website

Project Developed by: Amar Panigrahi